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GRAPHICS
2026
SINGAPORE

CT2Yarn: Yarn Level Reconstruction of Crochet from Computed Tomography

Chang Luo, Nobuyuki Umetani

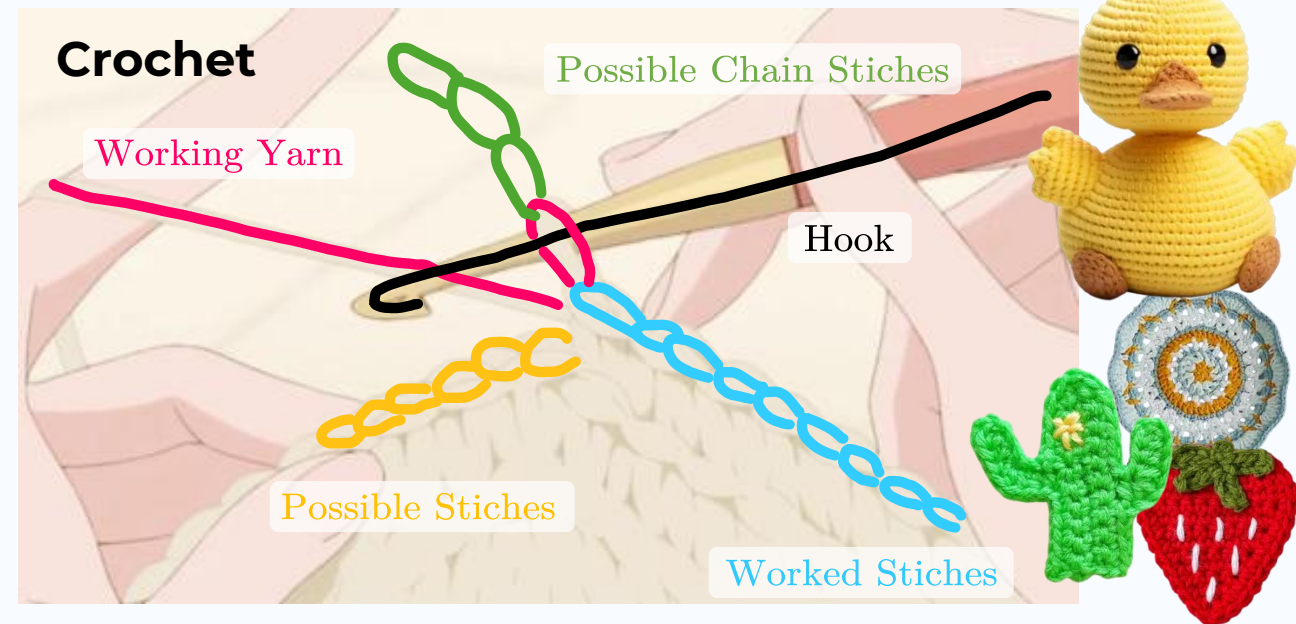
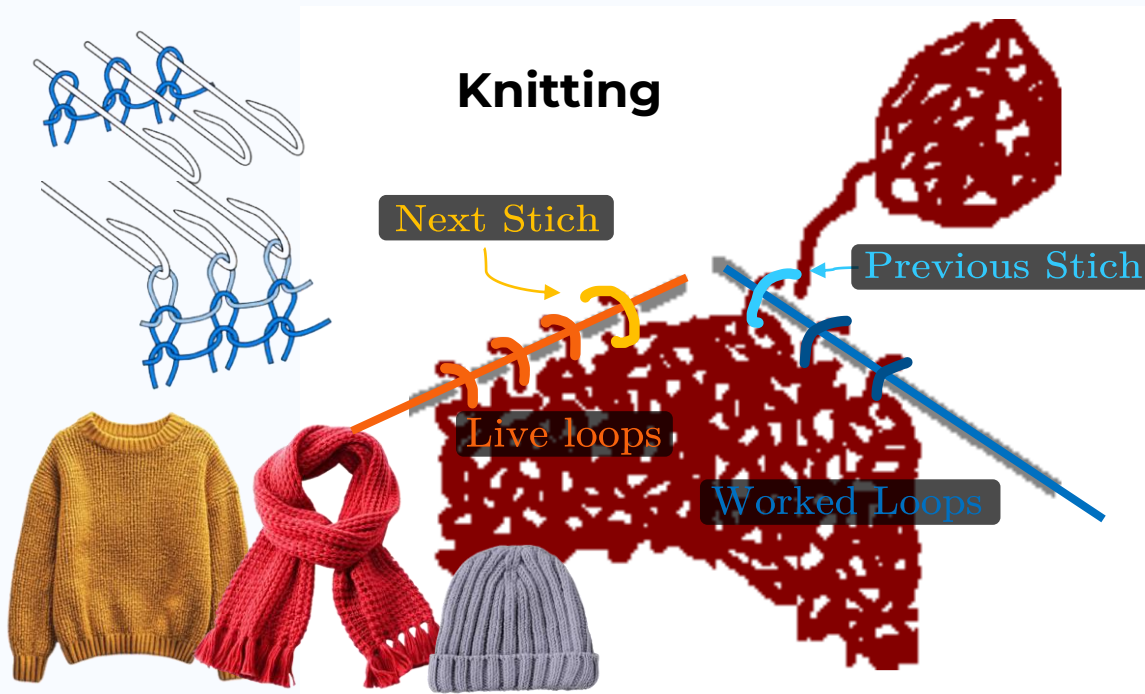
The University of Tokyo

October 2026 · Singapore



Background: What's Crochet?

- Distinct from Knitting, fabricated as a uniform 2D grid.
- Crochet is fabricated in 1D Line has more freedom.



- **Machine fabricable**, just put multiple hooks.
- **Hand-craft only**, due to the flexibility.
- Mainly for sweaters, scarfs, hats, etc.
- Mainly for amigurumi, flat patch decoration, etc.

Background: Why Computed Tomography?

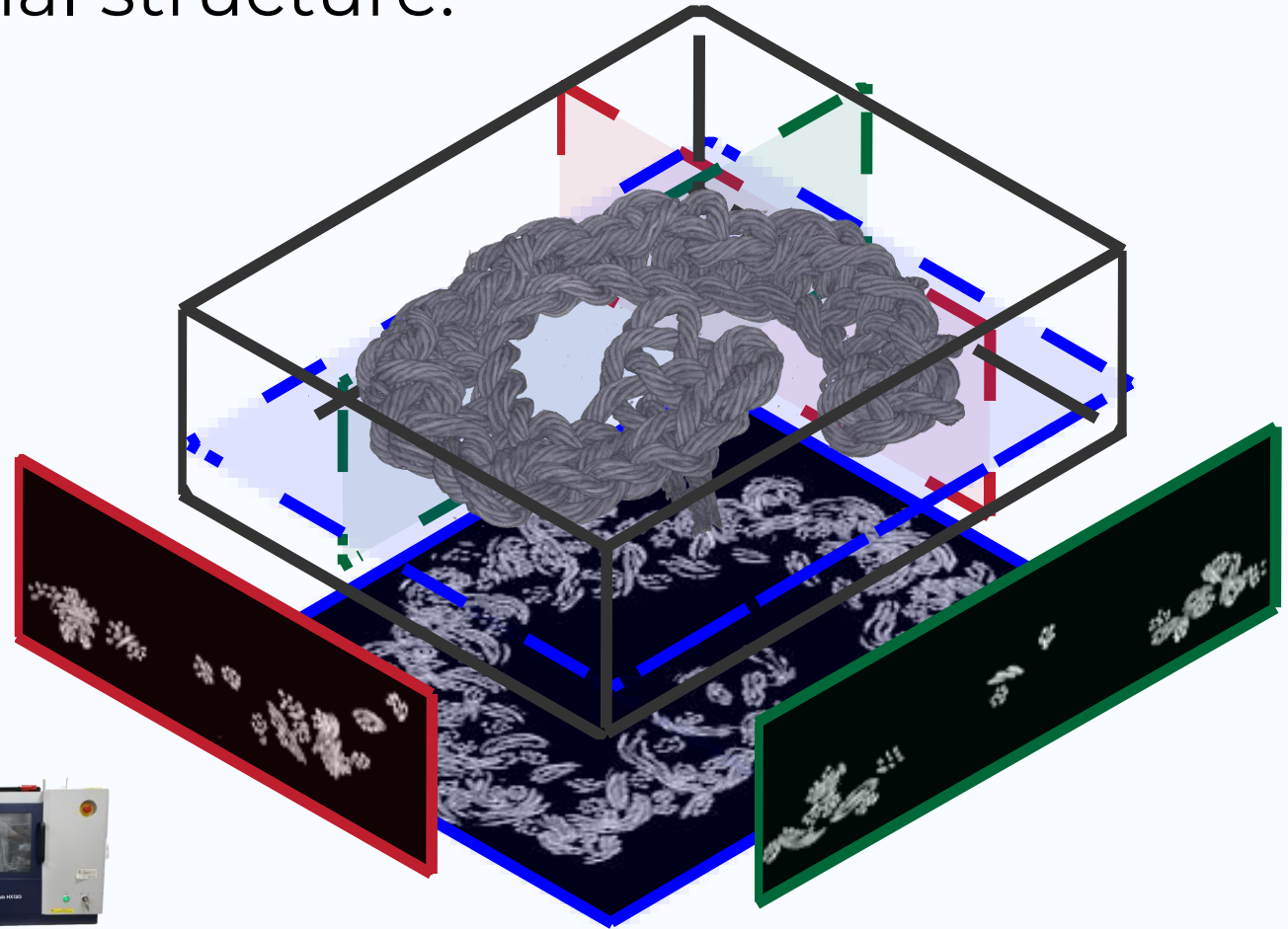
- CT Scan can see internal structure.



2D Image of Crochet



CT Scanner



CT Volume with slices

- Propose a method to **reconstruct the crochet**.



Crochet



CT Volume Data

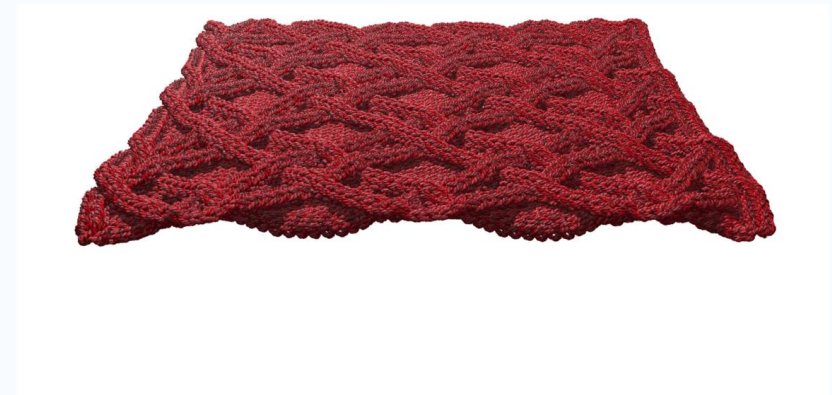


Centerline

- If we could reconstruct the yarn geometry structure, then...
 - More **realistic fiber-level rendering** of the crocheted fabrics.
 - **Physics Simulation** with mechanical properties.



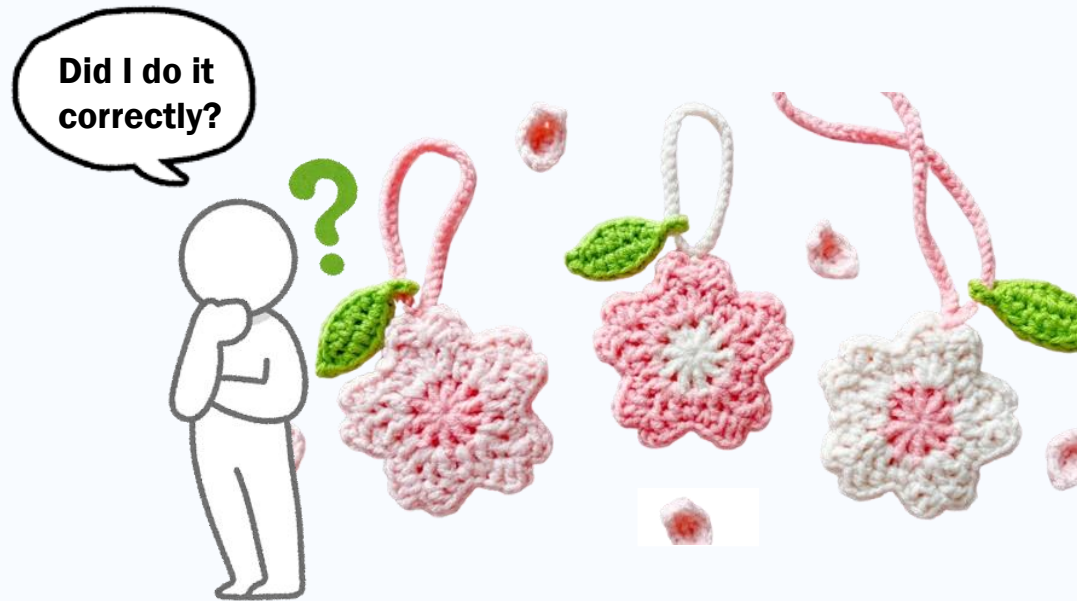
Physics Based Rendering



Physics Simulation

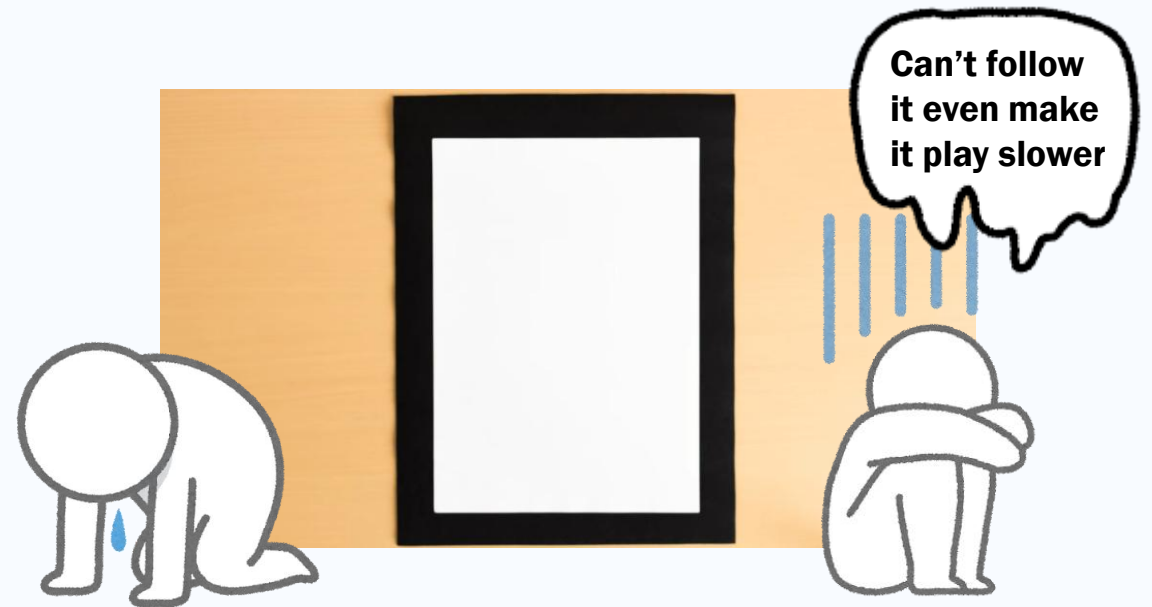
Motivation: Potential Applications

- Semantics reconstruction will help...
 - Recover the **unknown pattern**, or **validation** of the made crochet.
 - Stich by stich replay of the crochet process as **teaching tutorial**.



Bought crochets from somewhere

<https://www.youtube.com/watch?v=9ThrkgGhBVw>



Video Tutorials are hard to follow

<https://youtu.be/oAlGR-6nInI?si=vMMrchuMs75cvBd5>

Related Work: Knitting Reconstruction [Alexandre+19]

- Recover Knit Instruction from 2D Images.

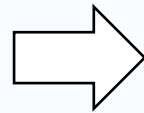


Limitation: Knitting is very organized

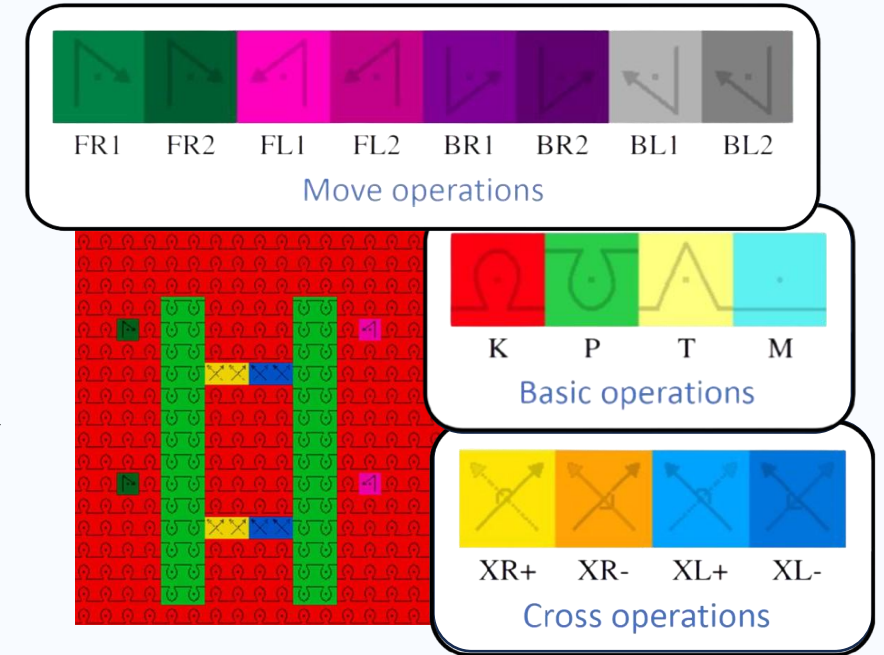
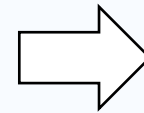
- **2D Image** is too hard for Crochet
- **No Uniform Grid** in Crochet



Capture



2D Image



Machine Knit Instruction

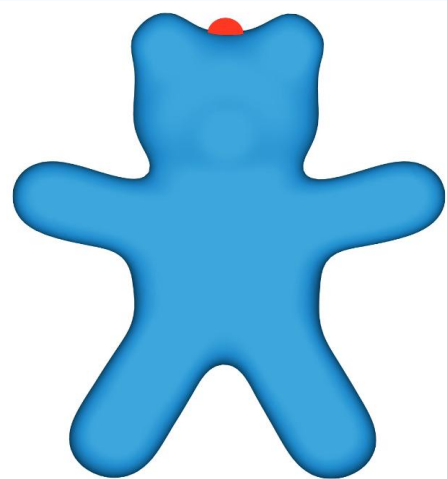
Related Work: Crochet Synthesis [Michal+22]

- Generate text crochet instruction for Closed Mesh.

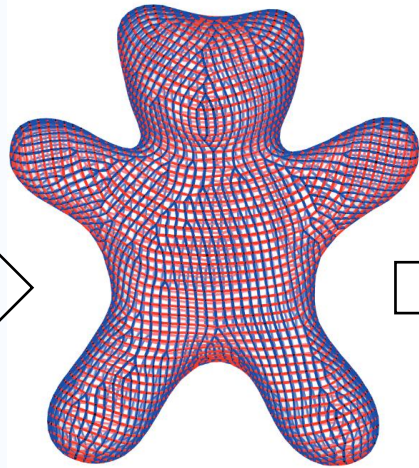
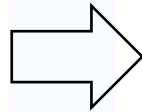


Limitation: Different Modelling Direction

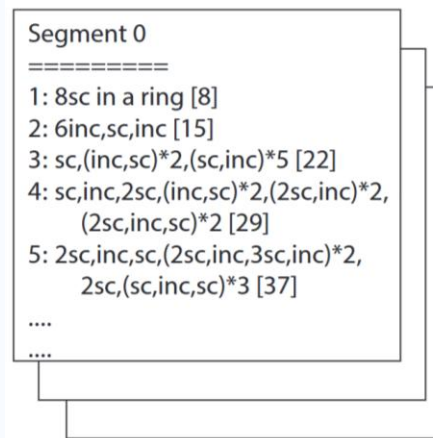
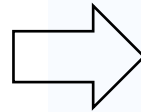
- This is **Forward Synthesis**
- Designed **only for Amigurumi**



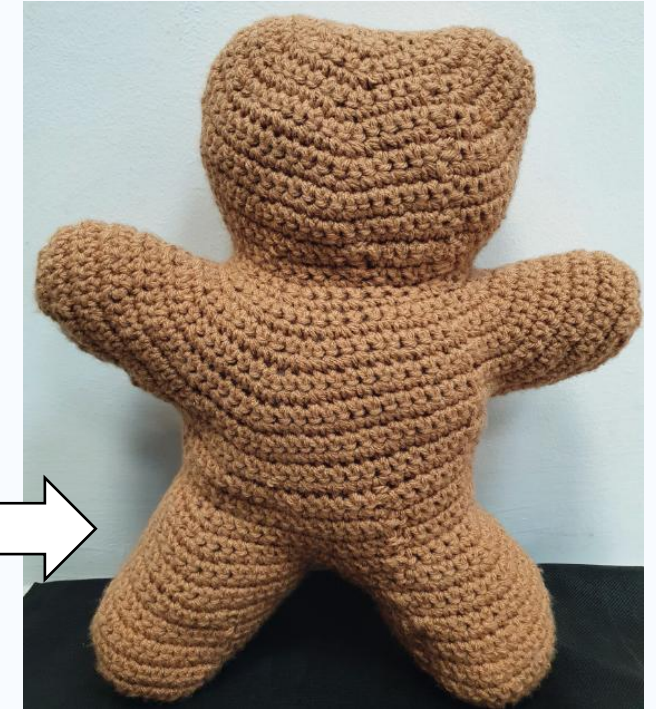
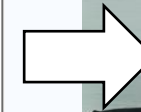
Input



Crochet Graph



Crochet Instruction



Hand-crafted Crochet

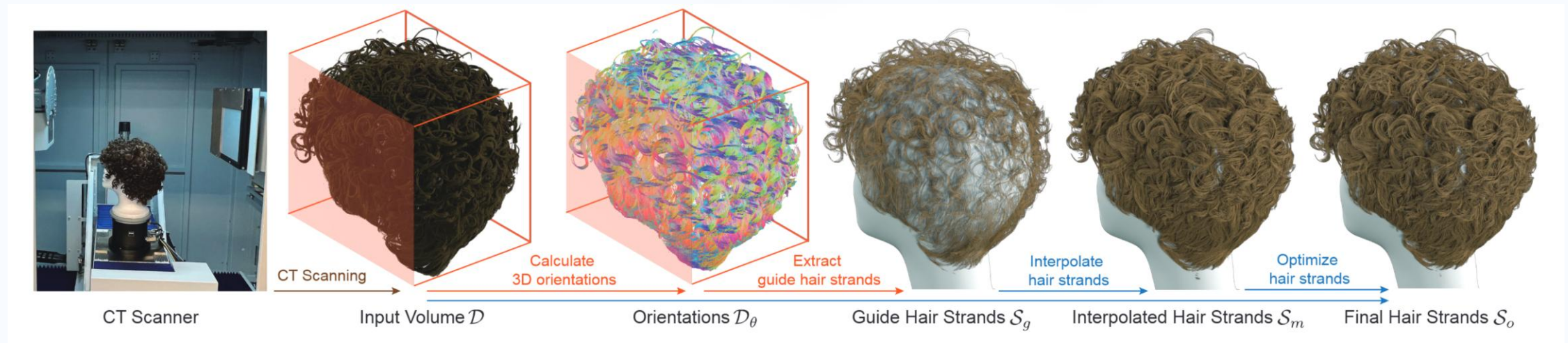
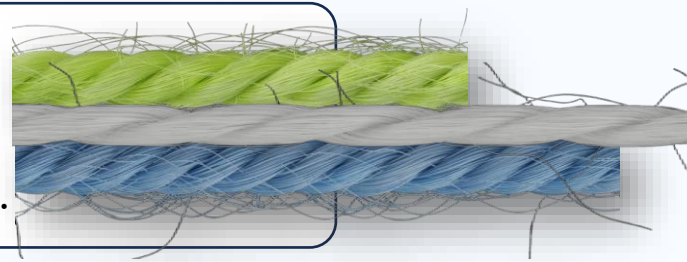
Related Work: CT Hair Reconstruction [Shen+23]

- 3D CT Volume Scan $\rightarrow$ Hair strands.

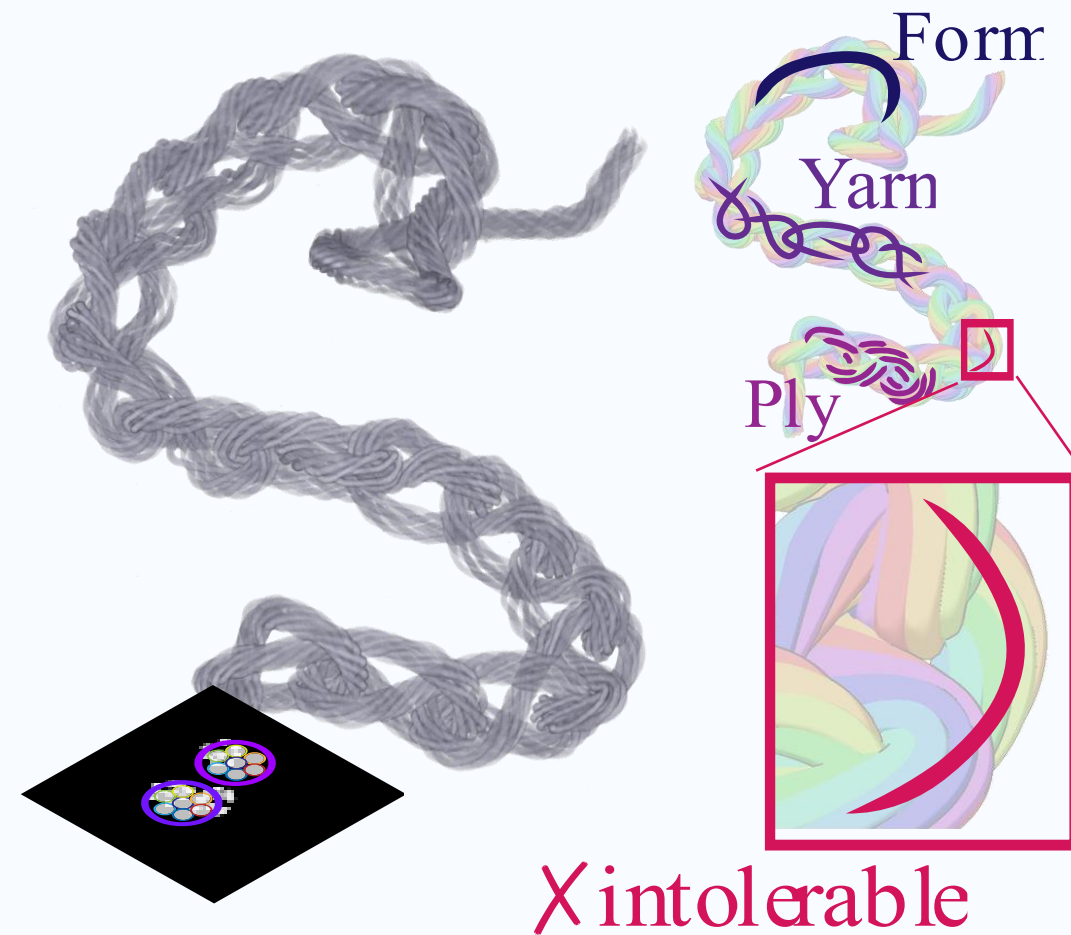
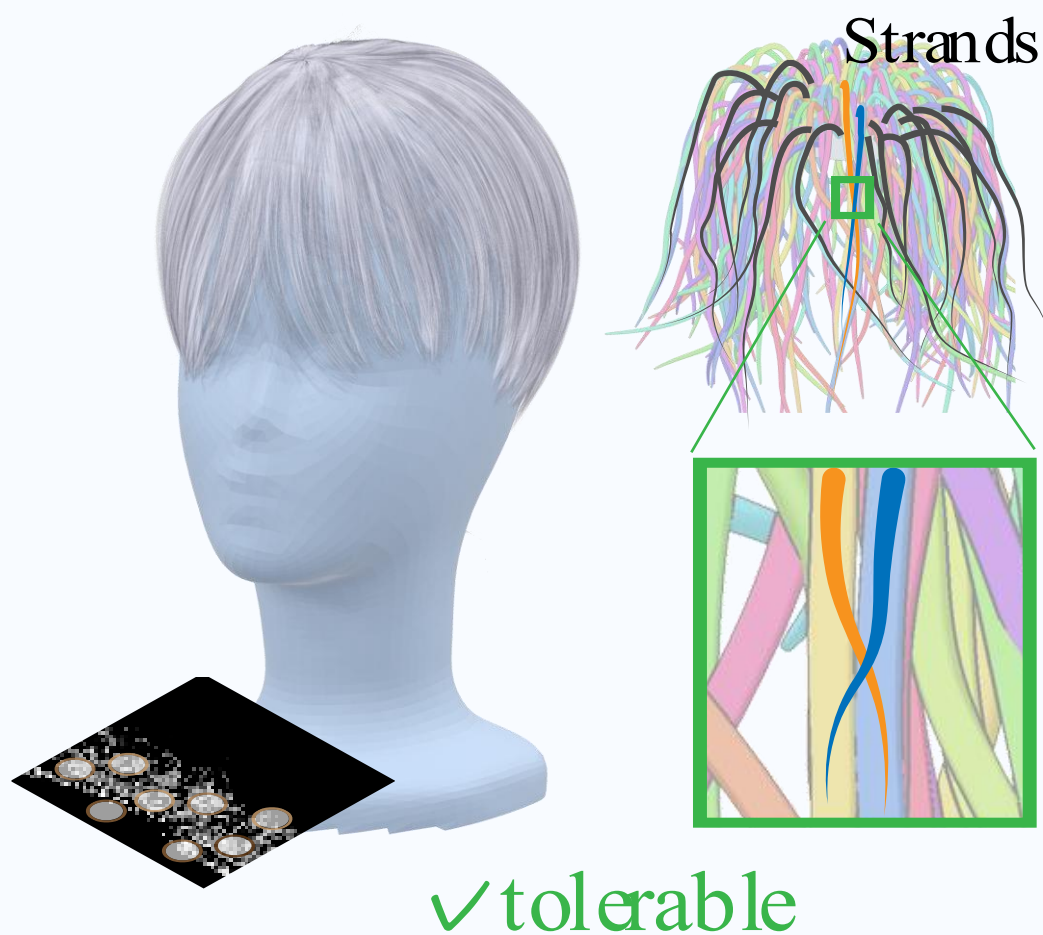


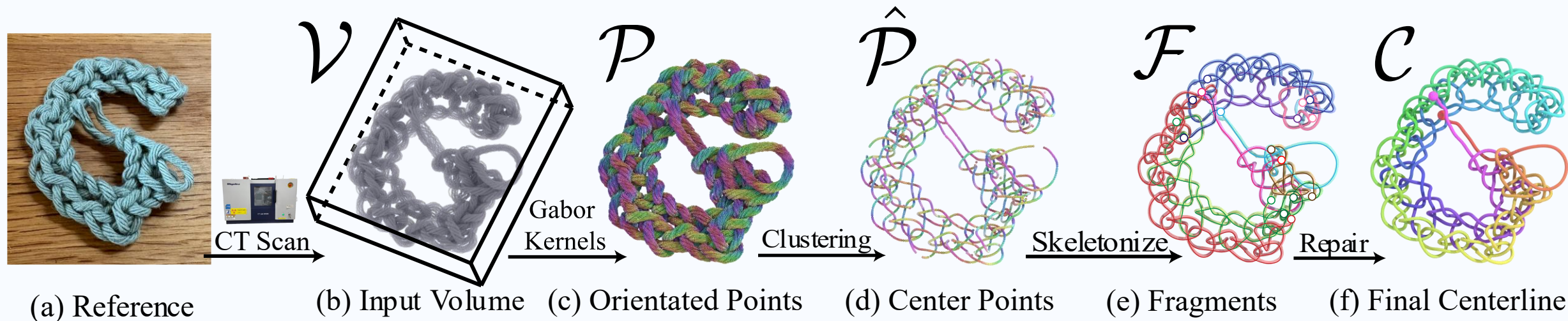
Limitation: Different Domain

- Hair is **different from yarn**.
- No accuracy** is required for hair.



- Hierarchy and Tolerance are different.

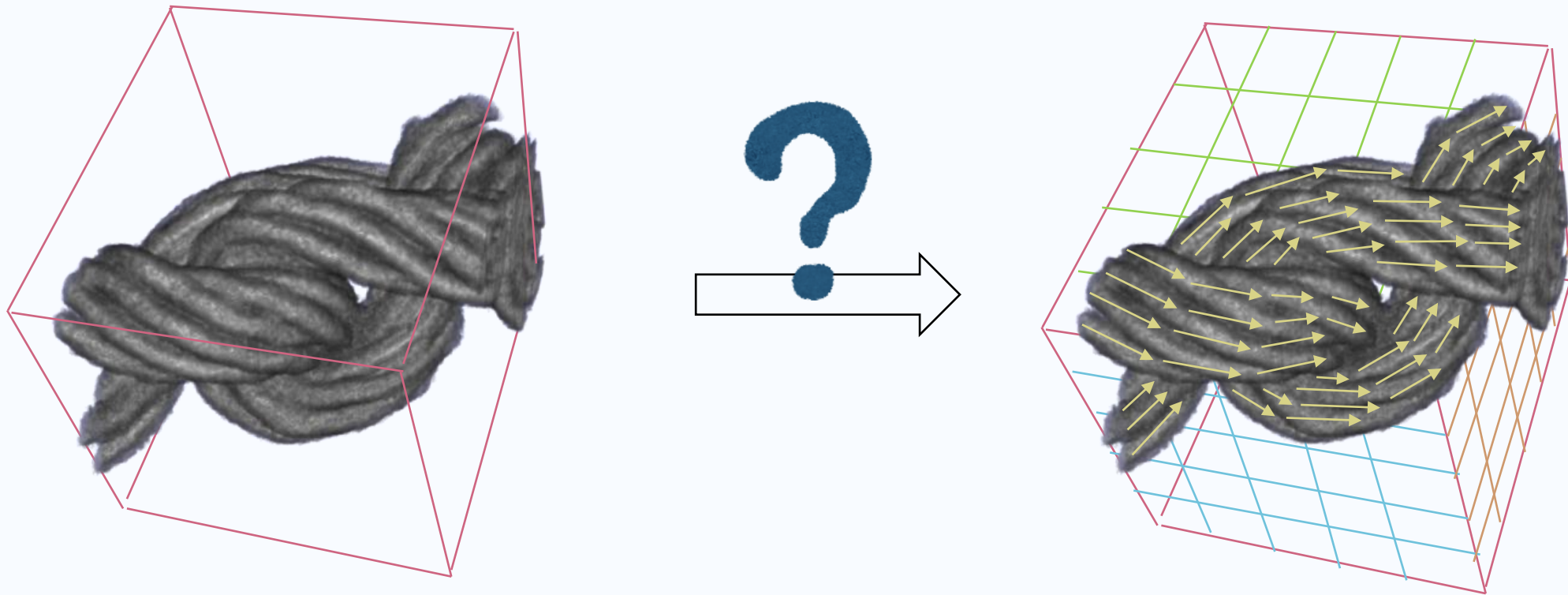




- (b) Volume $\mathcal{V}$: size in $H \times W \times D$, intensity value in each voxel.
- (c,d) Oriented Points $\mathcal{P}$ and Center Points $\hat{\mathcal{P}} : (X,Y,Z)$.
- (e) Fragments $\mathcal{F}$: local typology built from center points.
- (f) Centerline $\mathcal{C}$: global typology of the entire crochet yarn.

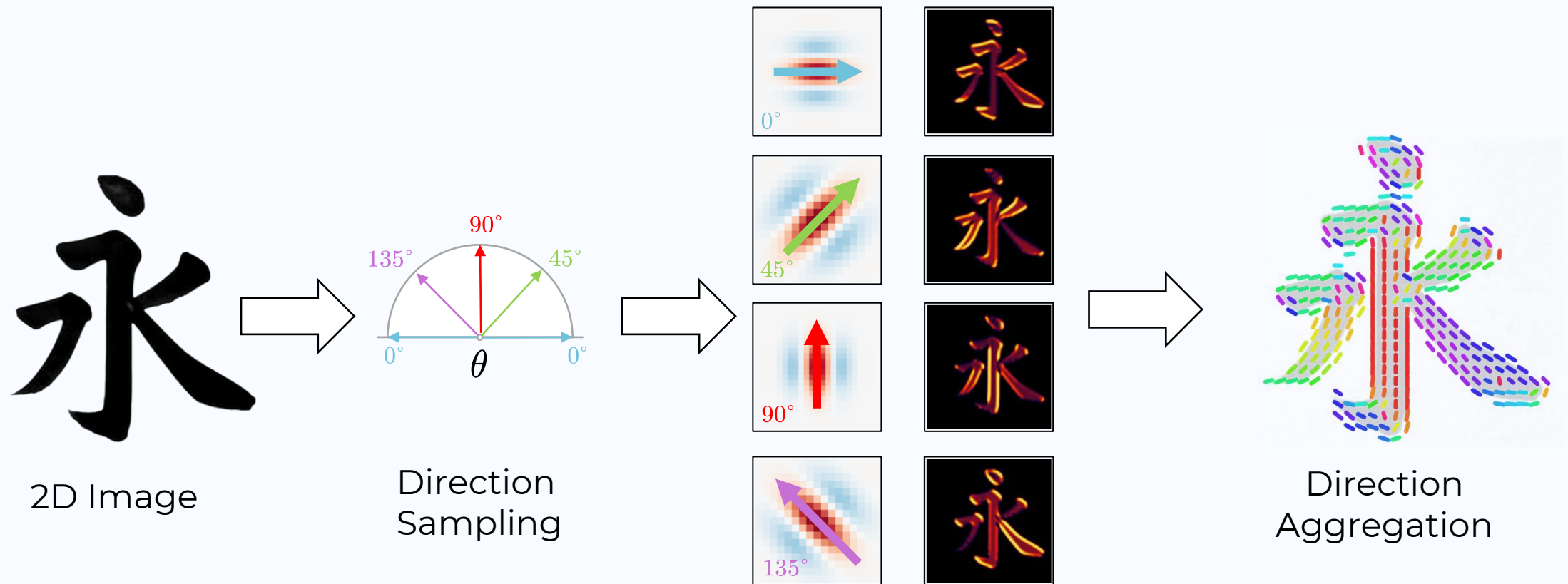
Method: Direction detection

- Given a 3D CT Volume, want to compute **direction per voxel**.



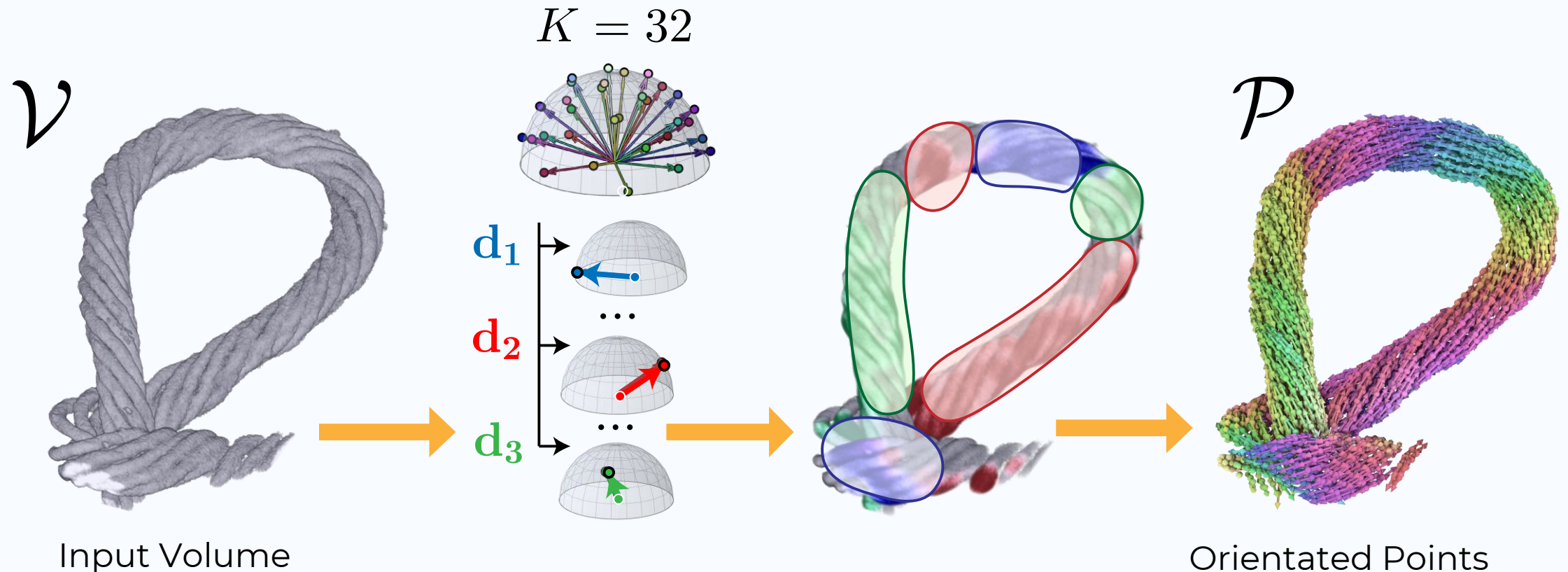
Method: Direction detection with Gabor kernels

- Gabor kernels could be used to detect directions.



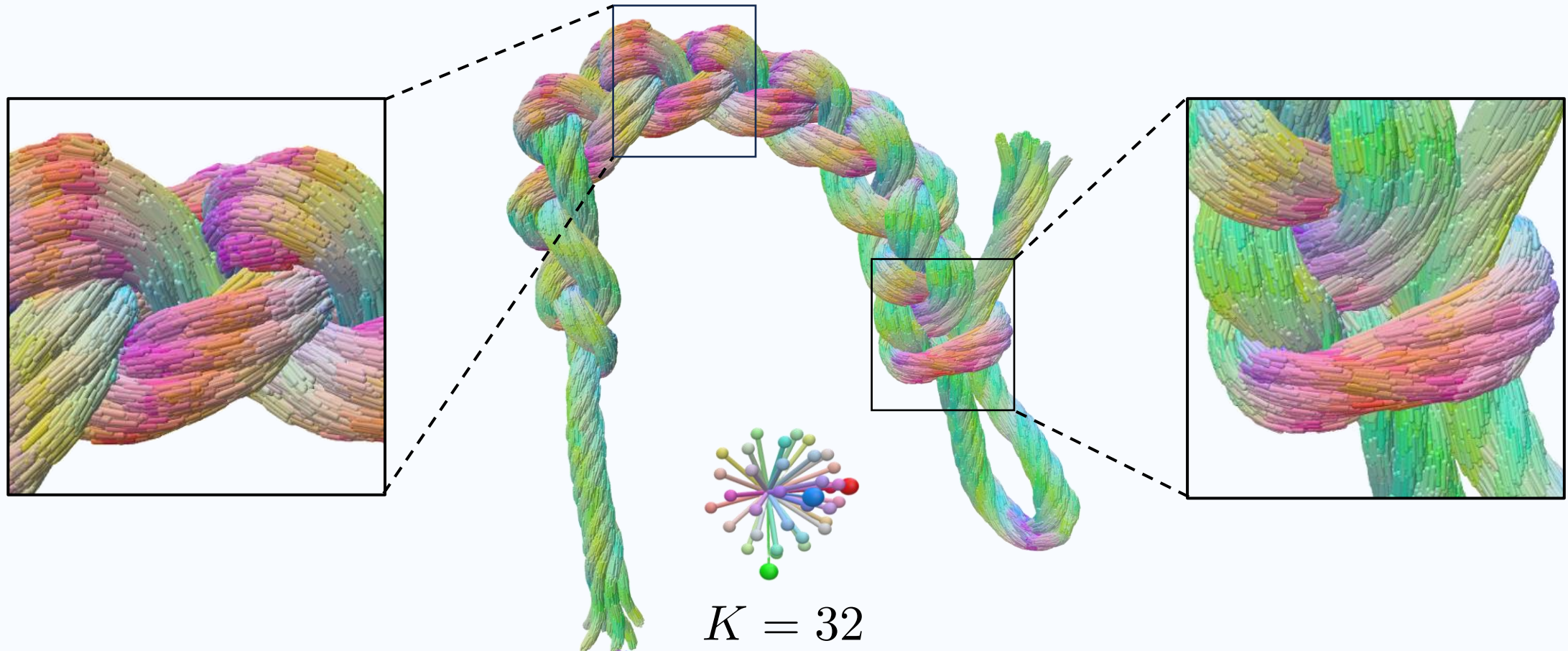
Method: Gabor in 3D

- $K = 32$ directions sampling on a hemi-sphere.
- For each direction calculate the response in Volume $\mathcal{V}$.



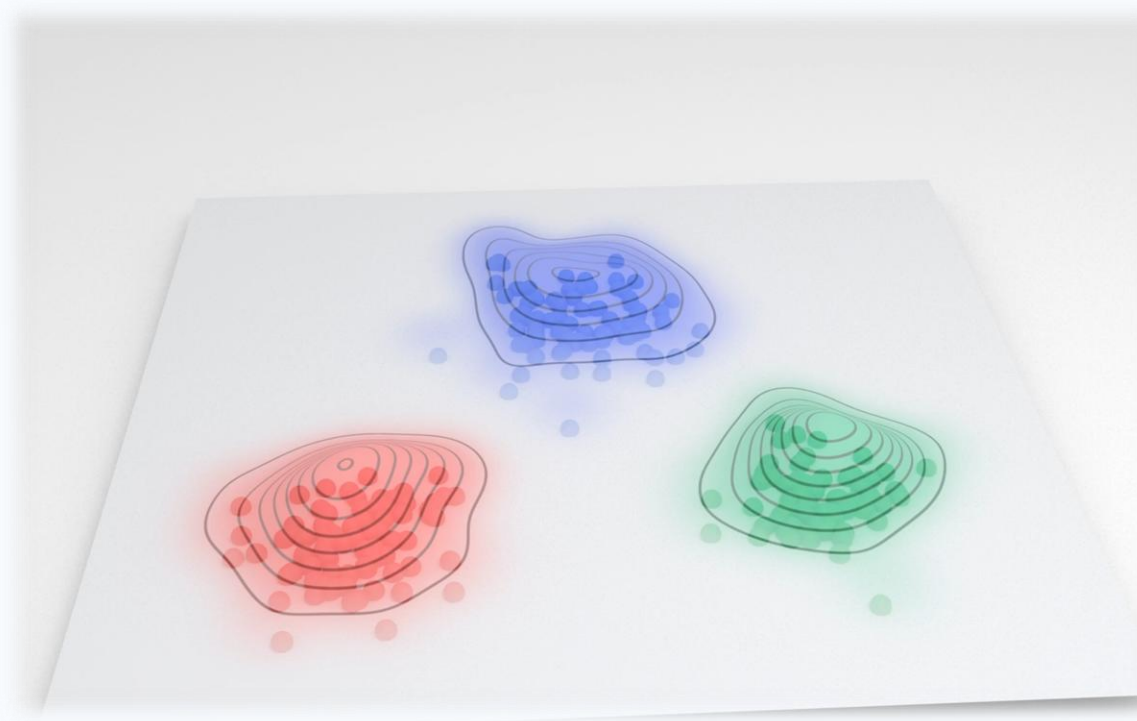
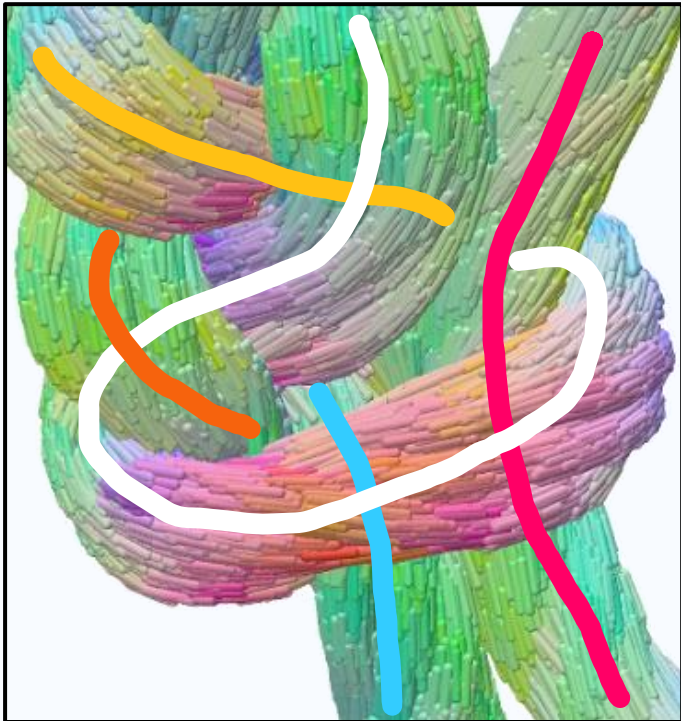
Method: Direction detection

- Directions are extracted per voxel.



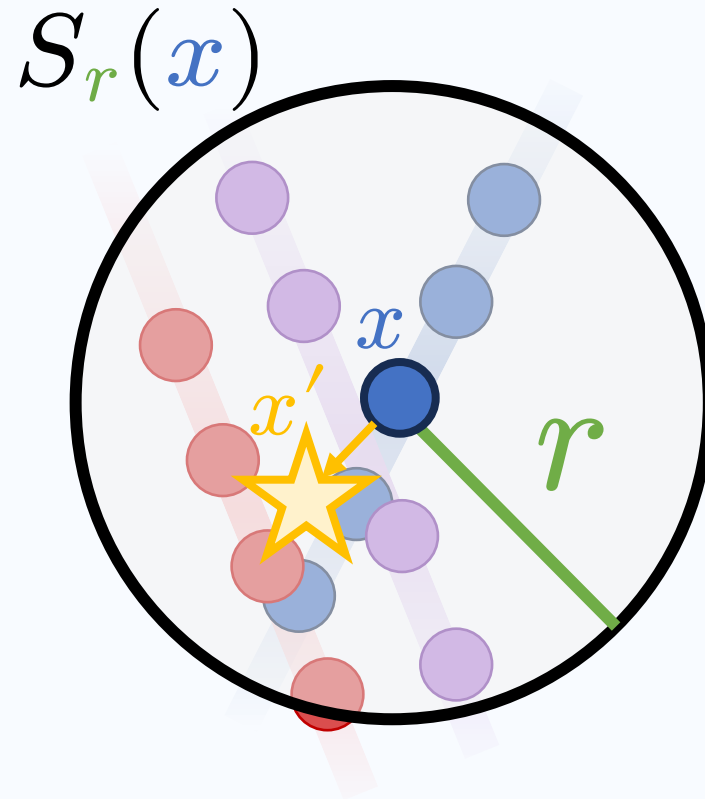
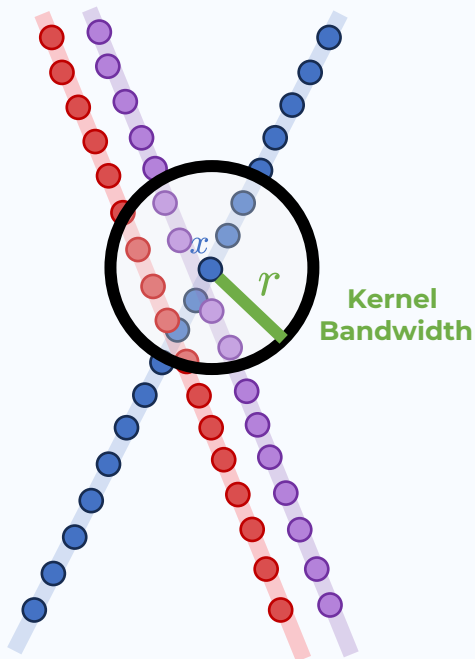
Method: Centerline Extraction

- Where is the skeleton (centerline) of the yarn?
- Mean Shift is an algorithm for clustering.

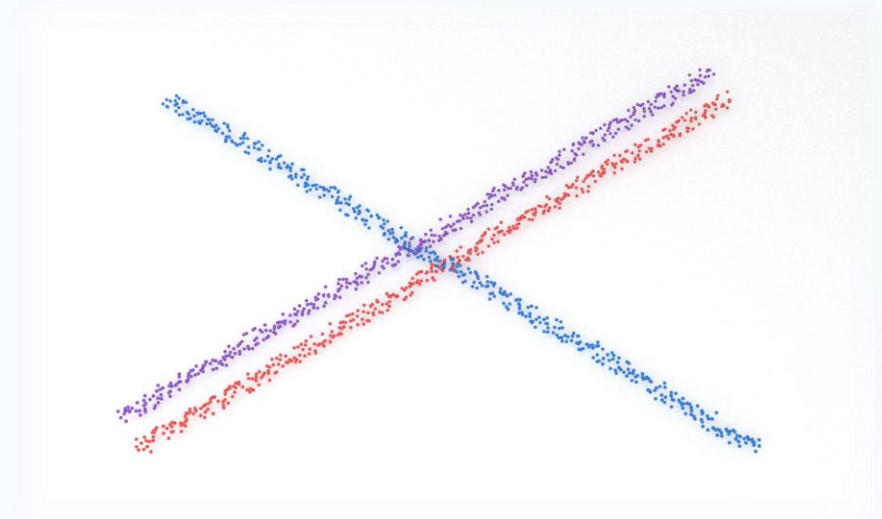


Method: Centerline Extraction by Mean-Shift

- Mean-Shift push points to their local centers iteratively.

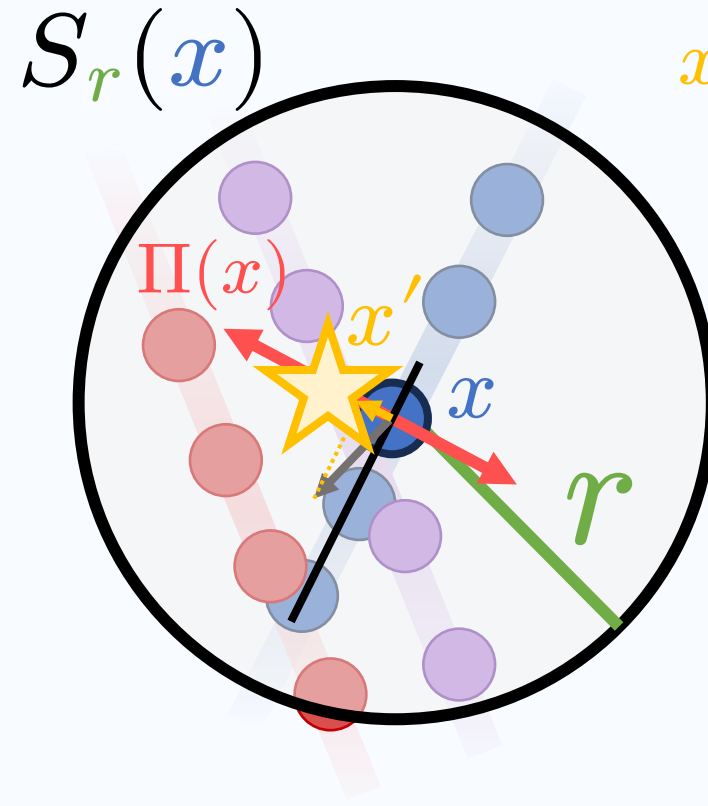
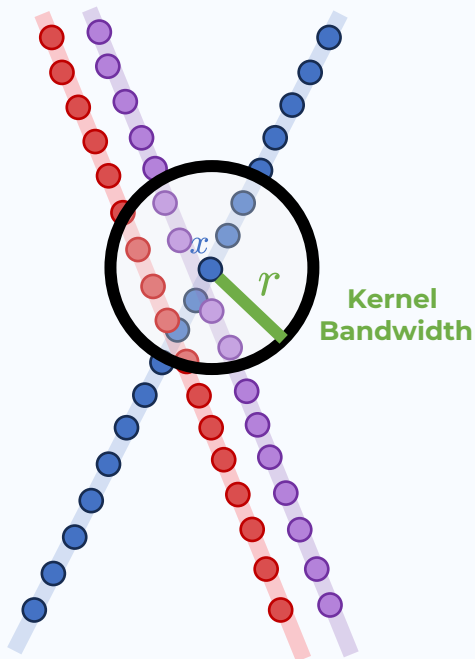


$$x' = \frac{1}{|S_r(x)|} \sum_{x_i \in S_r(x)} x_i$$

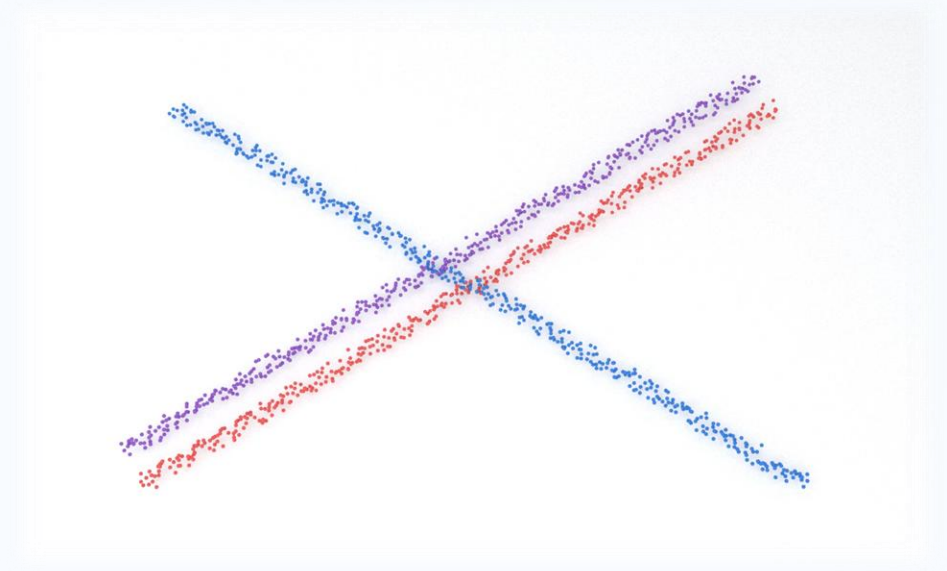


Method: Centerline Extraction by Mean-Shift

- Project the shift to perpendicular direction.

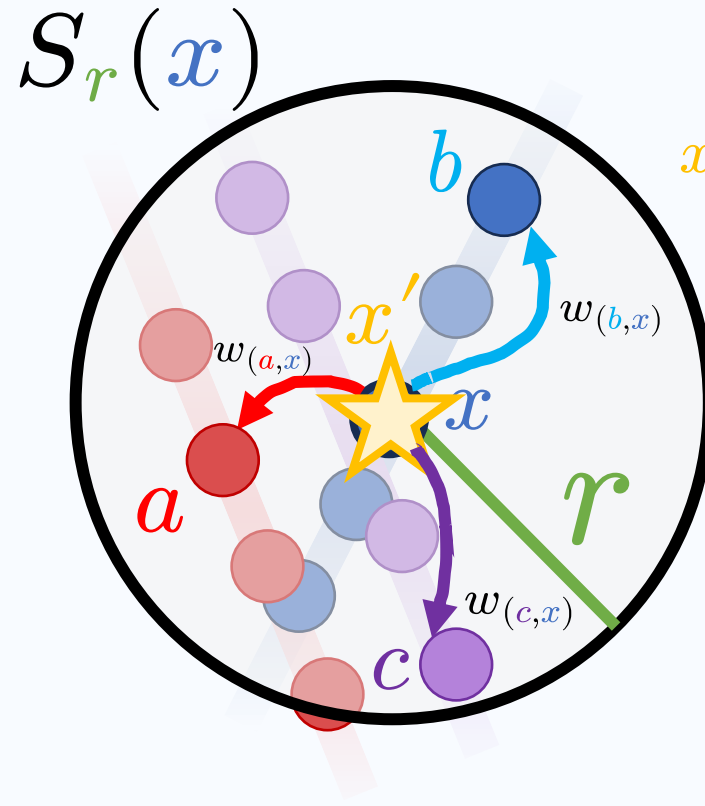
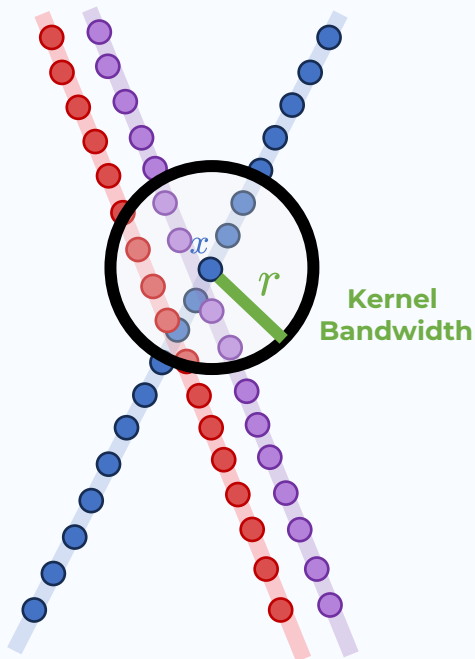


$$x' = \text{Proj}_{\Pi(x)} \left(\frac{1}{|S_r(x)|} \sum_{x_i \in S_r(x)} x_i \right)$$



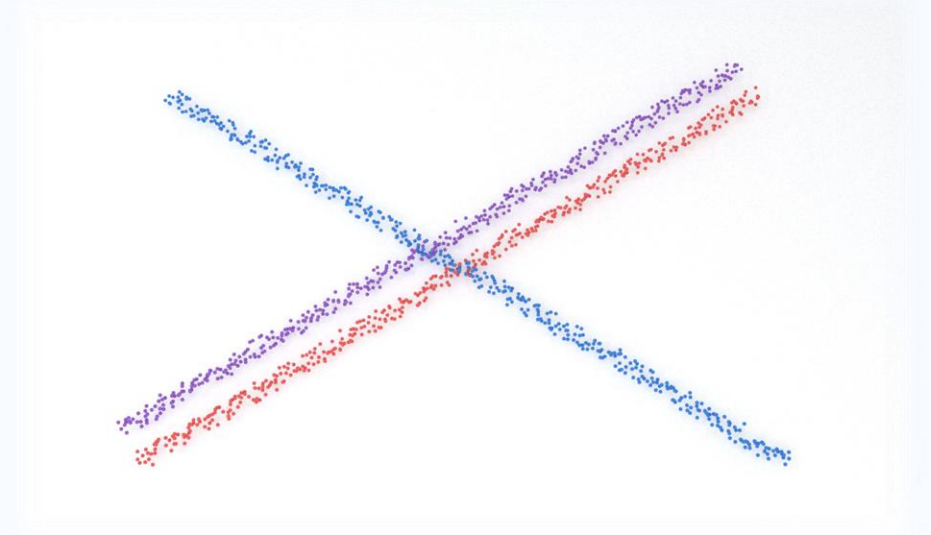
Method: Centerline Extraction by Mean-Shift

- Weighting with cosine similarity.



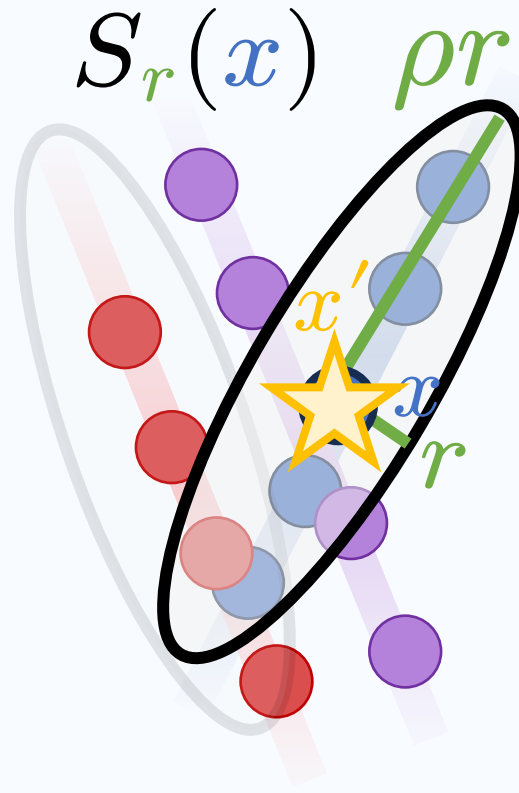
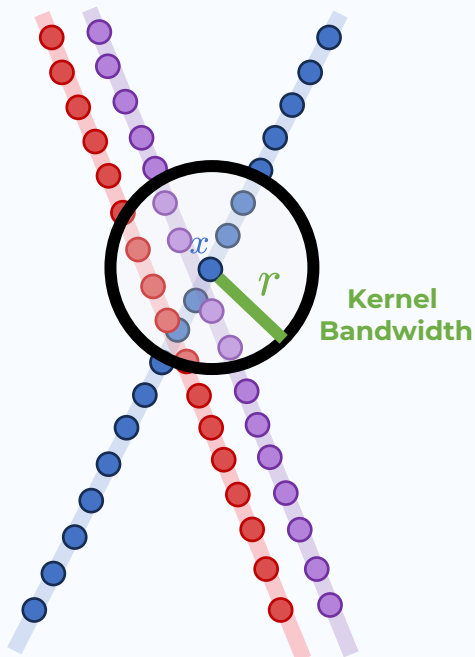
$$w(x_i, x) = \cos^\alpha(x_i, x)$$

$$x' = \text{Proj}_{\Pi(x)} \left(\frac{1}{\left| \sum_{x_i \in S_r(x)} w_i \right|} \sum_{x_i \in S_r(x)} w_i x_i \right)$$



Method: Centerline Extraction by Mean-Shift

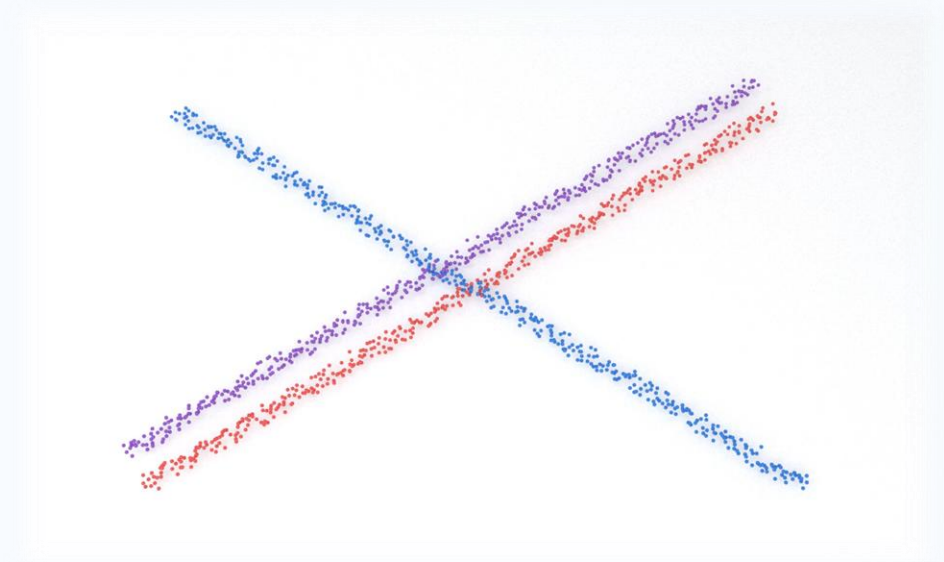
- Anisotropic kernel shape!



$$\sigma_{perp} = r \quad \sigma_{tan} = \rho r$$

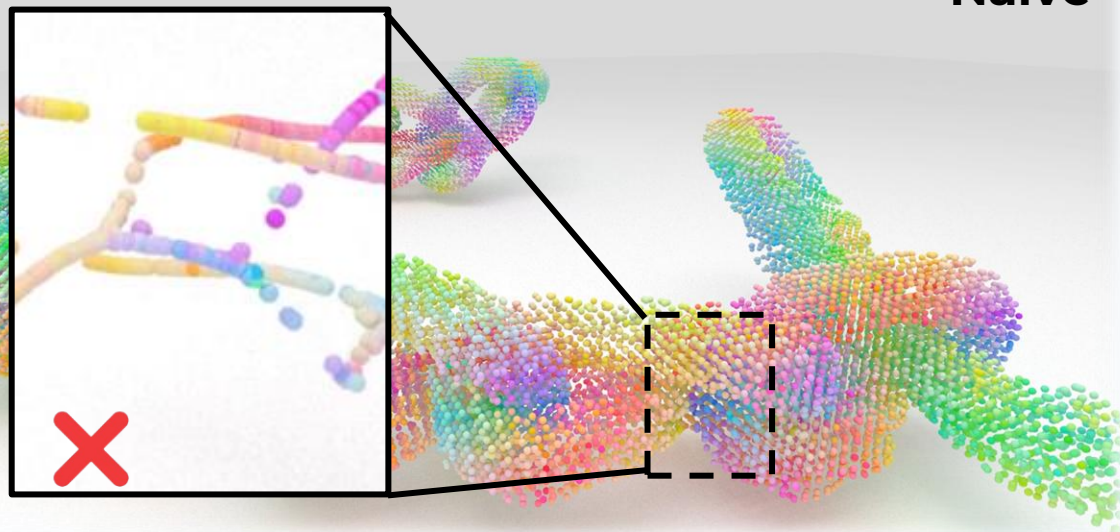
$$w(x_i, x) = \exp \left[-\frac{1}{2} \left(\frac{\delta_{tan}^2}{\sigma_{tan}^2} + \frac{\delta_{perp}^2}{\sigma_{perp}^2} \right) \right] \cos^\alpha(x_i, x)$$

$$x' = \text{Proj}_{\Pi(x)} \left(\frac{1}{\left| \sum_{x_i \in S_r(x)} w_i \right|} \sum_{x_i \in S_r(x)} w_i x_i \right)$$

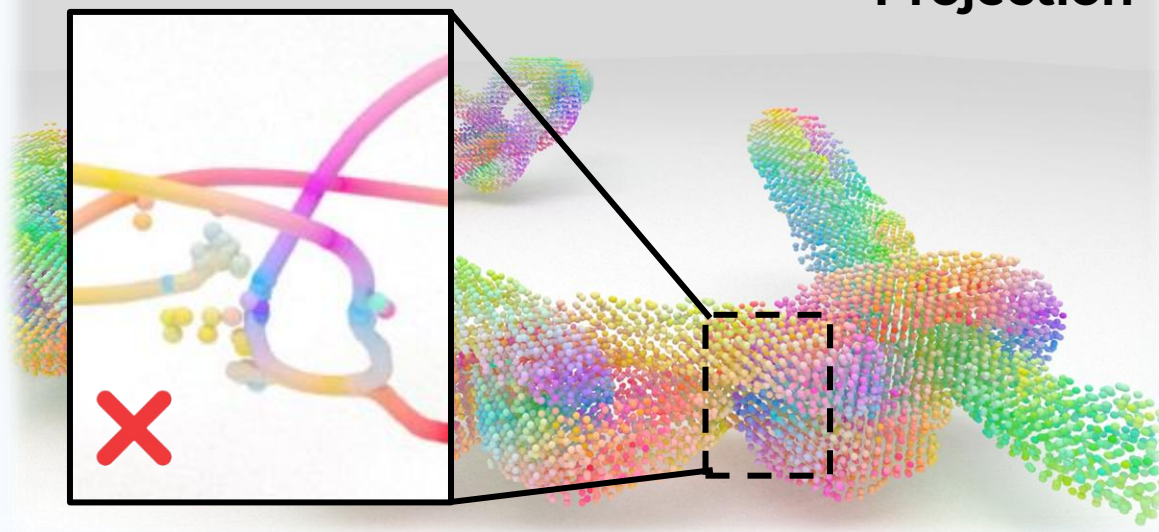


Method: Centerline Extraction by Mean-Shift

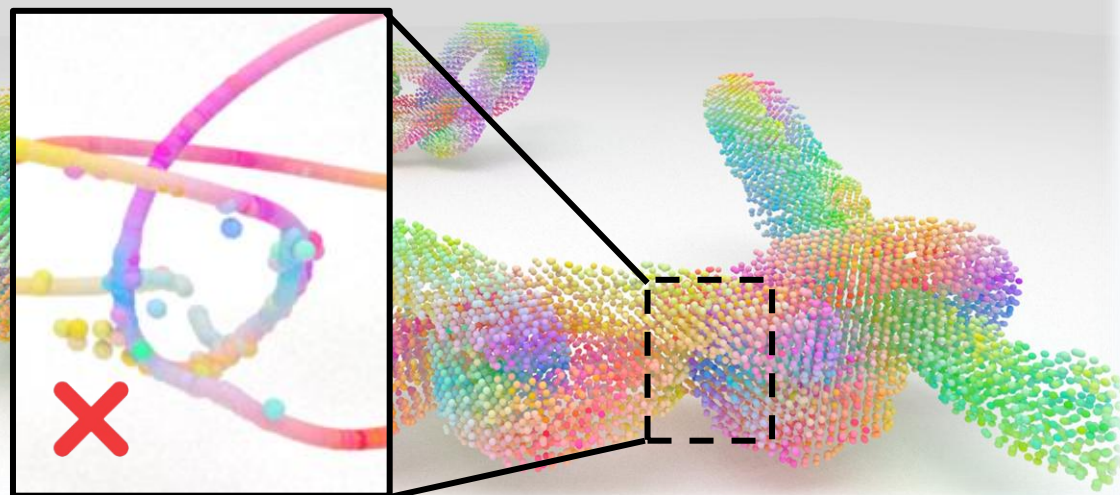
Naive



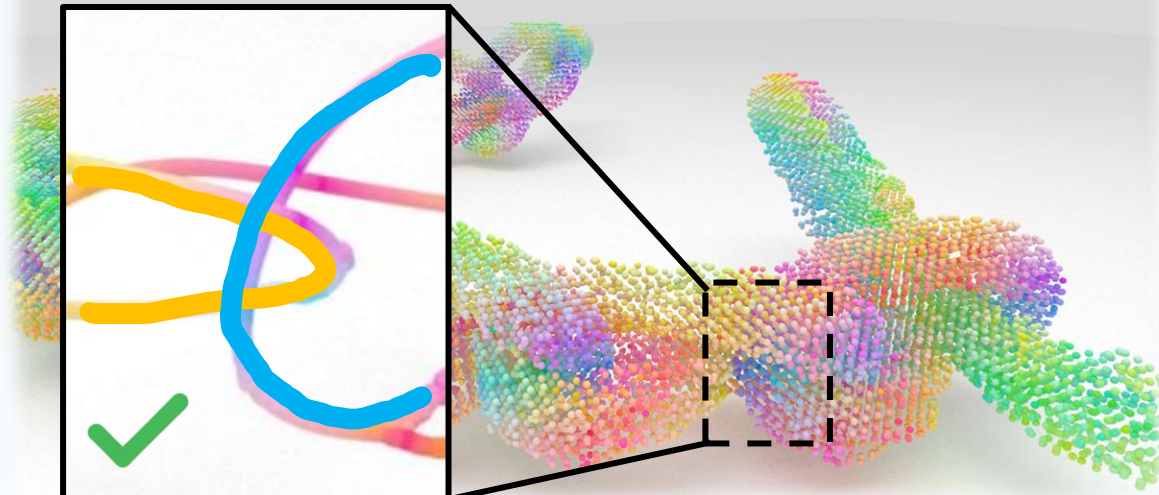
+ Projection



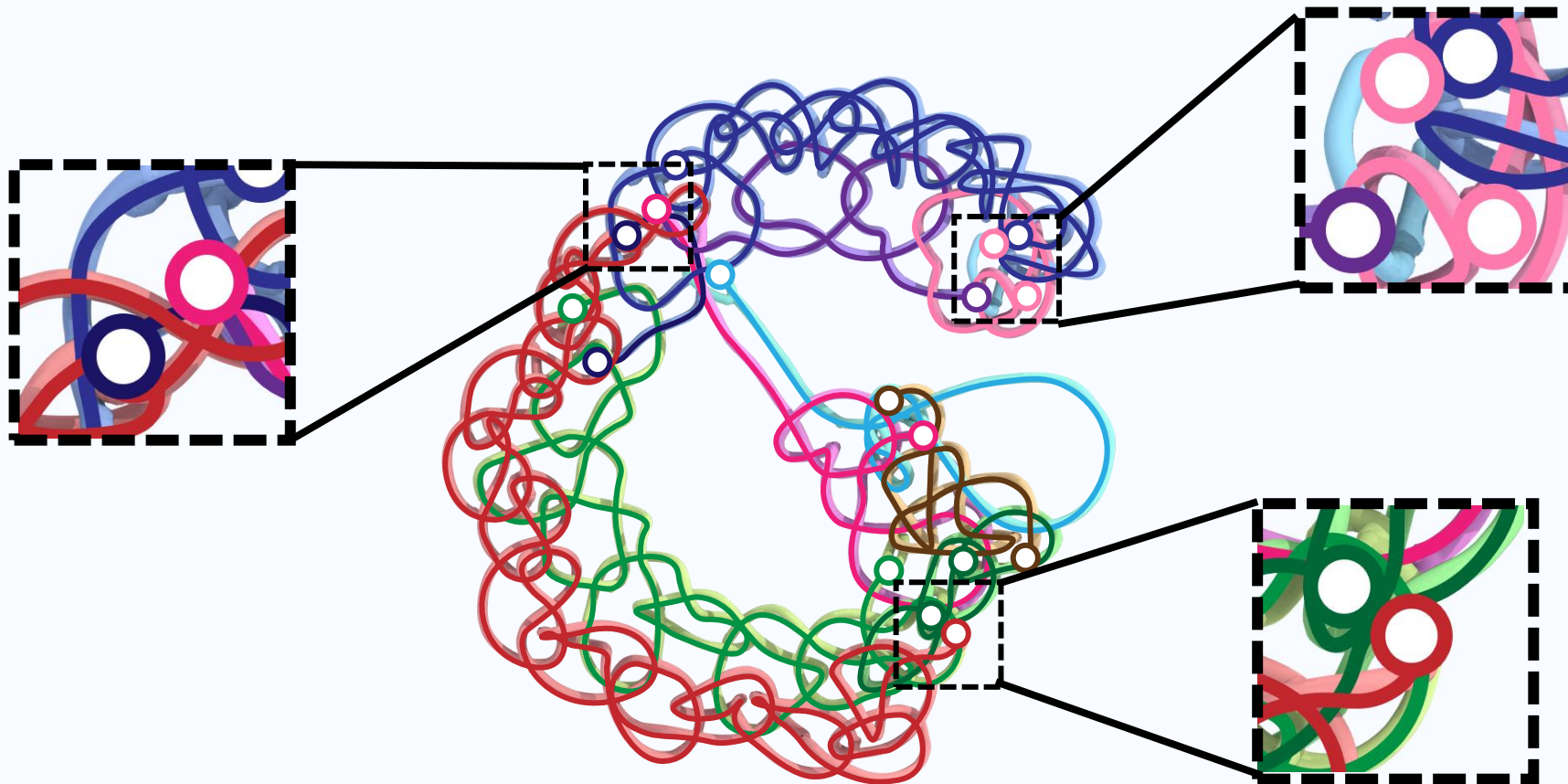
+ Projection + Direction



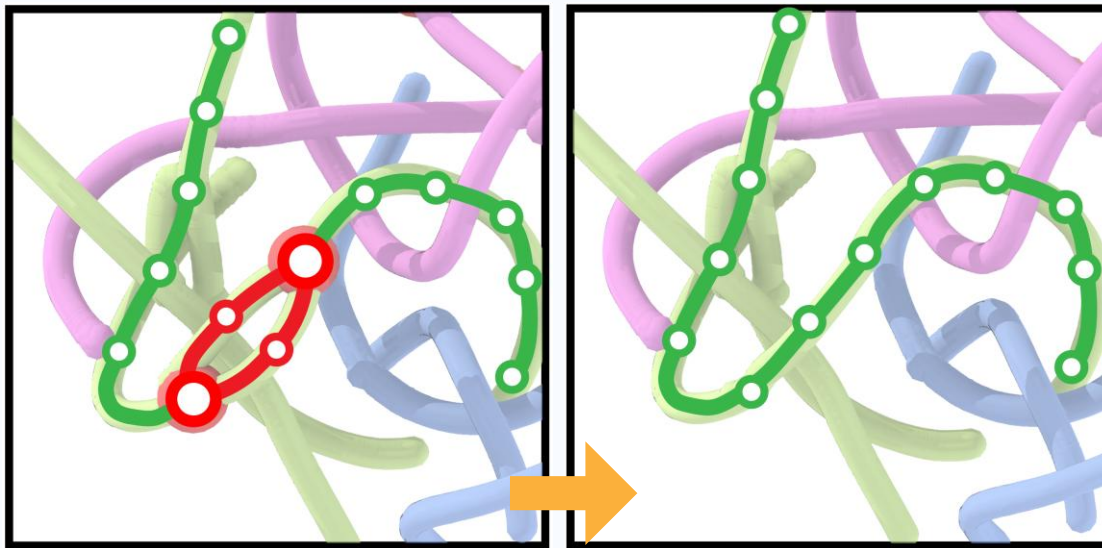
+ Projection + Direction + Anisotropic



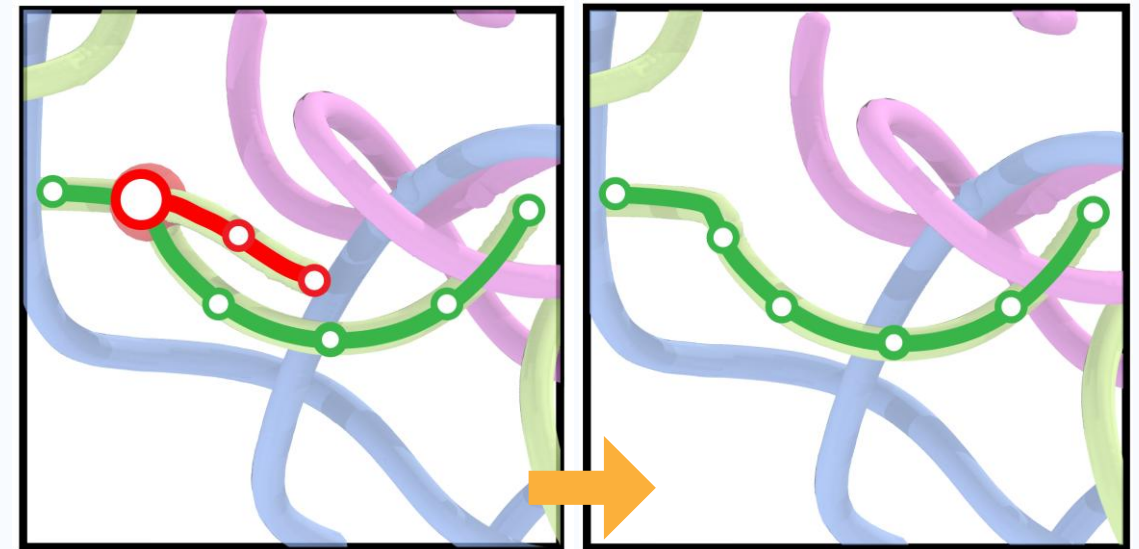
- **Minimum Spanning Tree** (MST) builds fragments with connection radius given by user.



- **Automatic** and manual tools to build global topology.



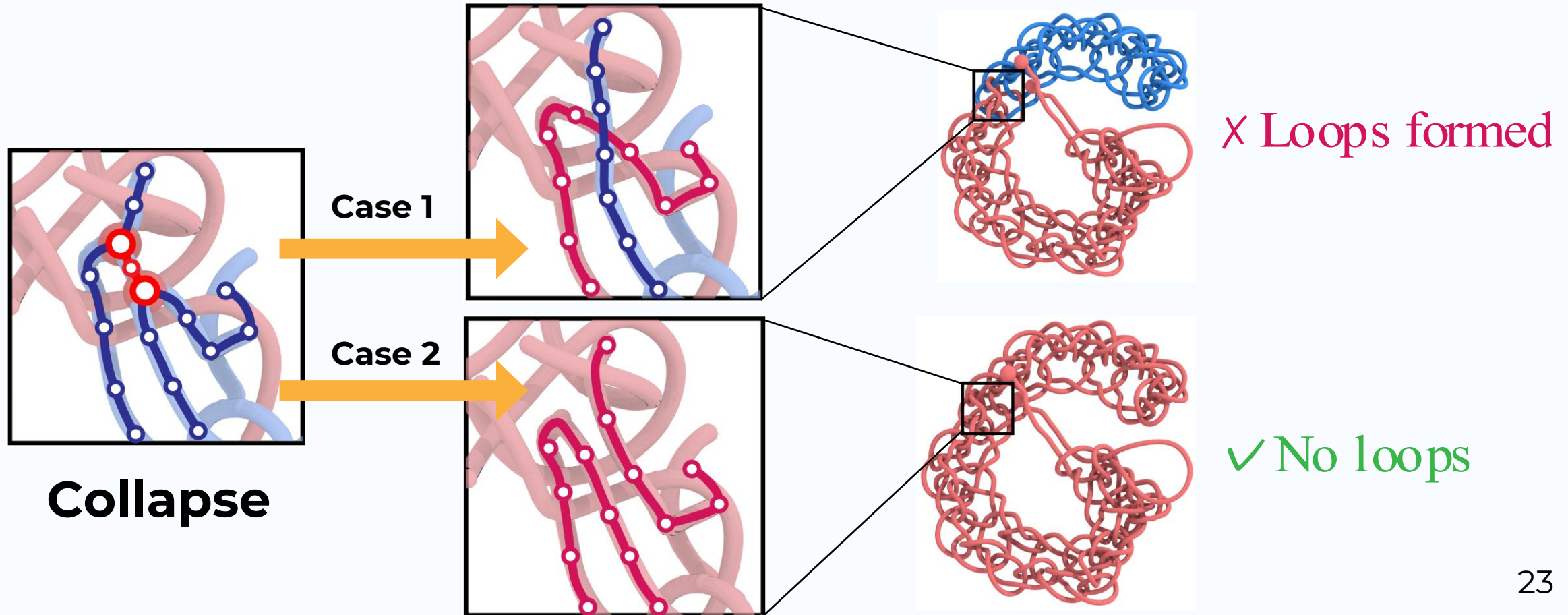
Ring



Twig

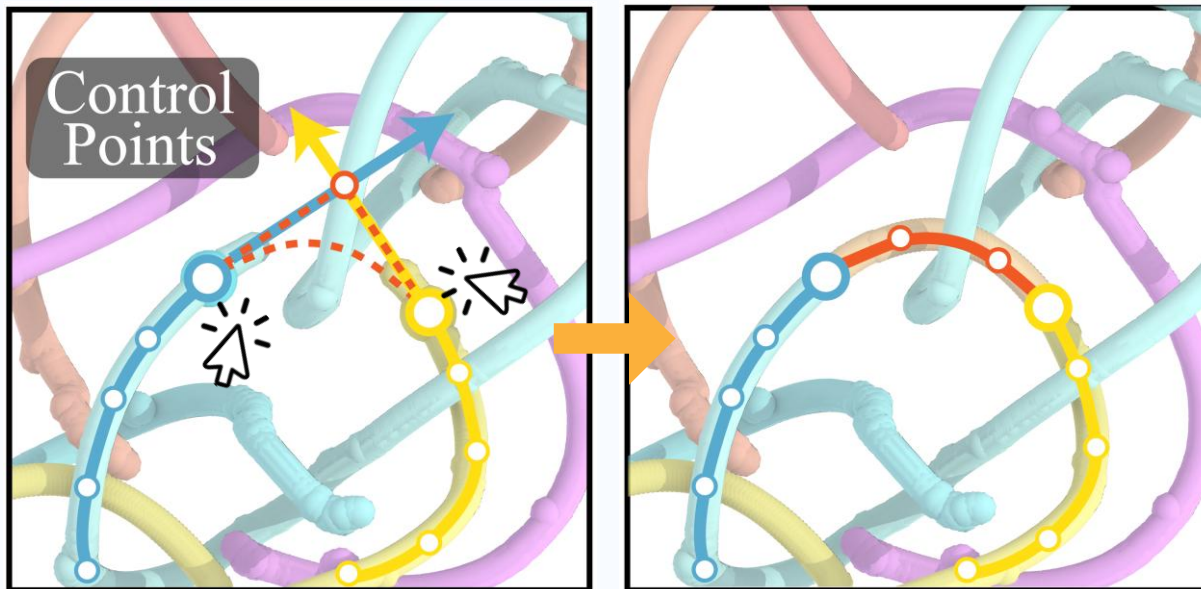
Method: Automatic Repairing

- Automatic and manual tools to build global topology.

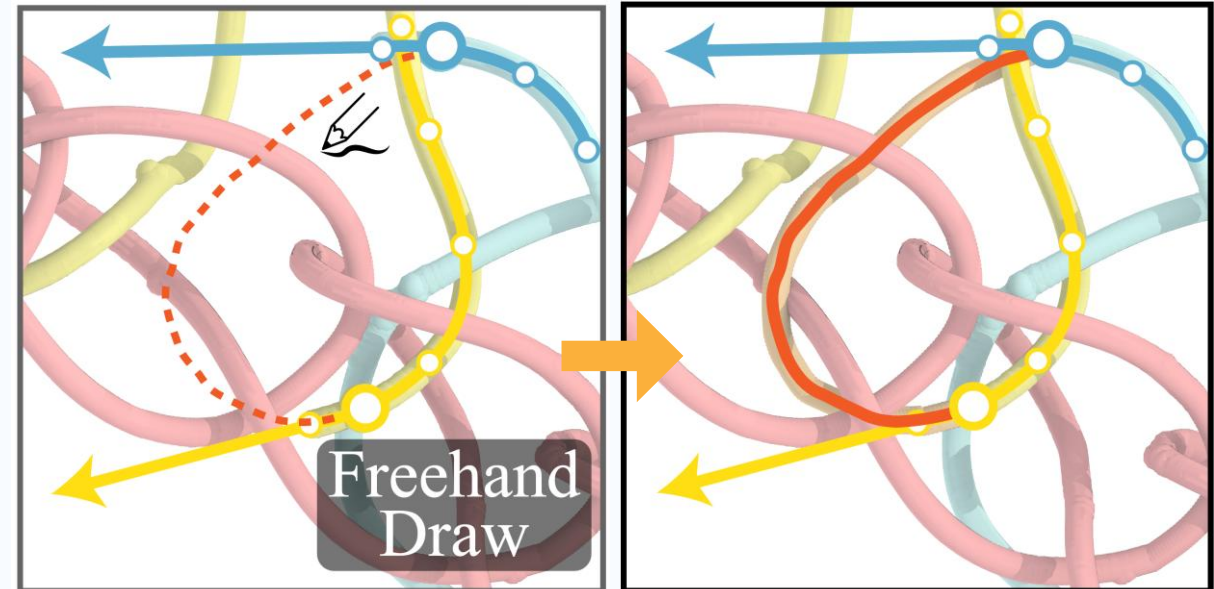


Method: Manual Repairing

- Automatic and manual tools to build global topology.



Bezier



Freehand

Results: Qualitative Evaluation

Color coded fragments

Red – Manual | Green – Auto

Reference

Chain C



Chain S



Chain O



Results: Qualitative Evaluation

Color coded fragments

Red – Manual | Green – Auto

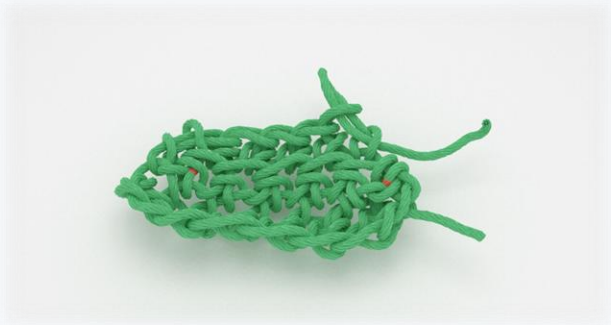
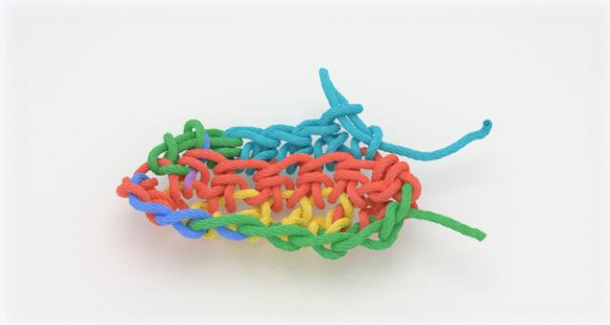
Reference



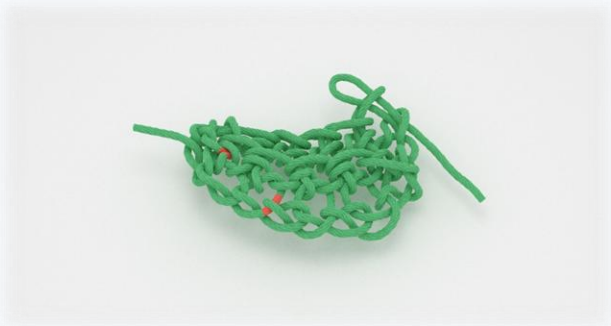
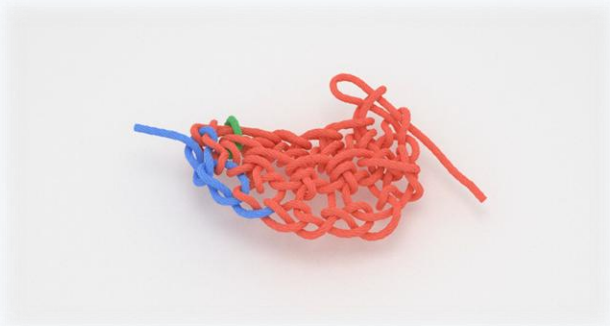
Patch 1



Patch 2



Patch 3



Results: Qualitative Evaluation

Color coded fragments

Red – Manual | Green – Auto

Reference

Letter P

Number 2

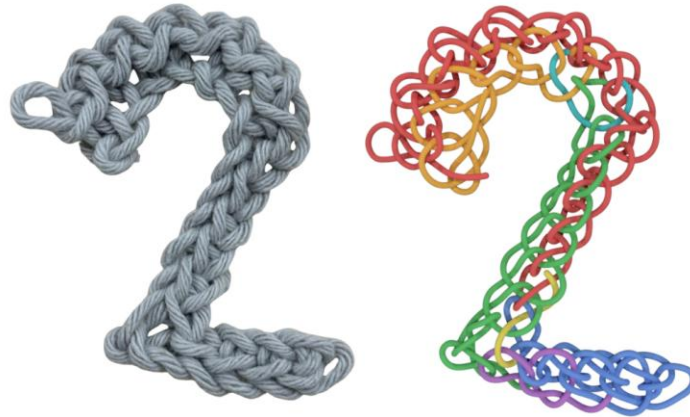
Letter G

Number 6

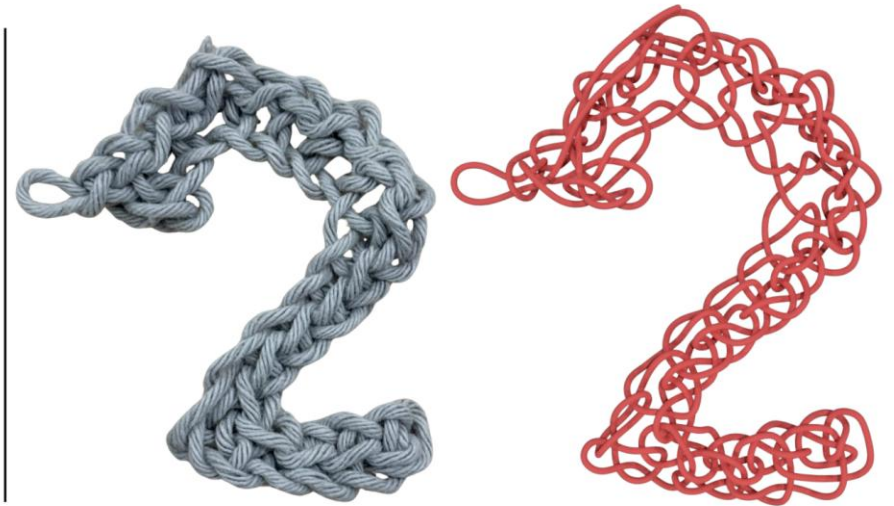
- Crochet sample stills need to be loose and simple.
- When tightly crocheted, manual interventions is necessary.



Tight crochet



Normal crochet



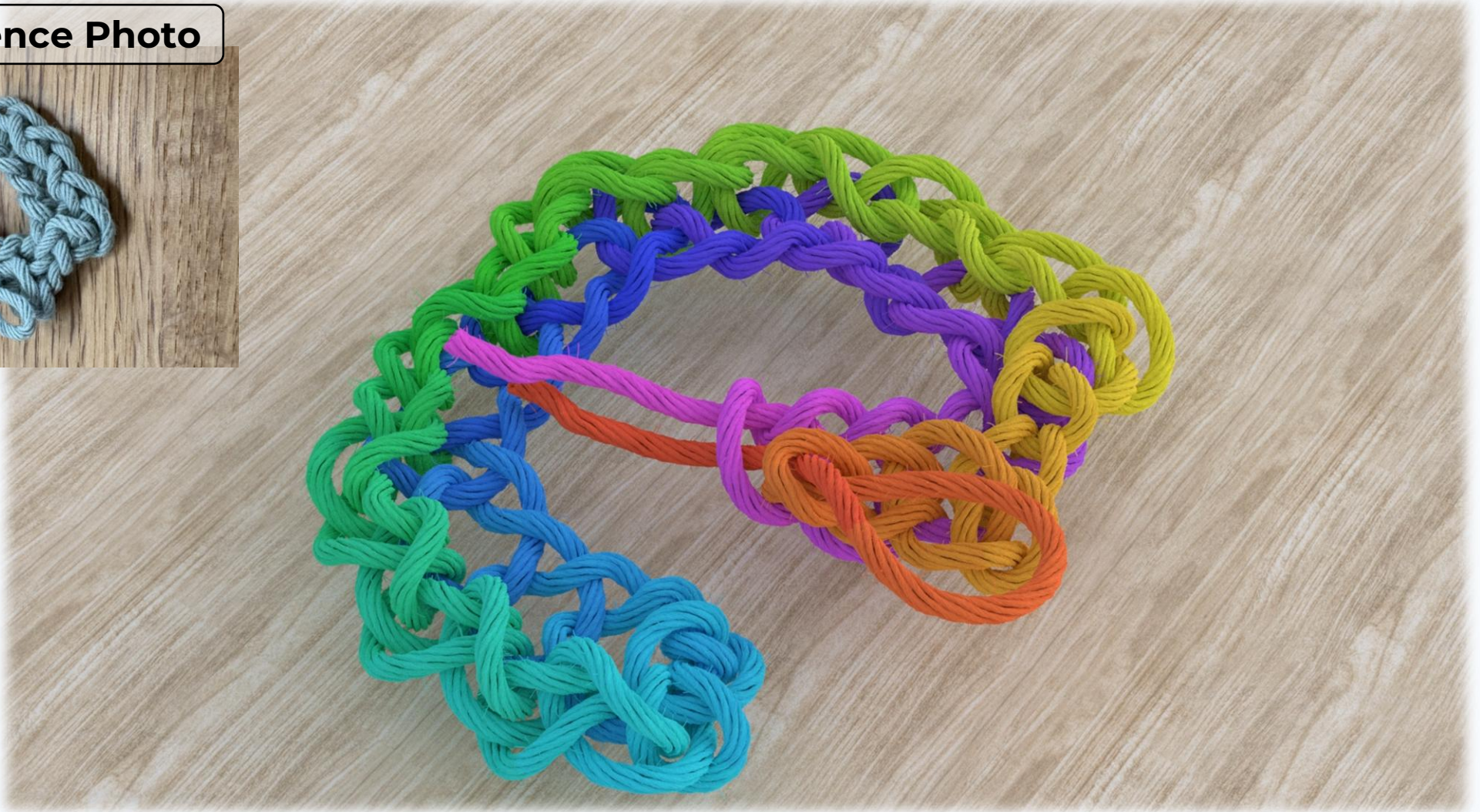
Loose crochet

Applications: Rendering



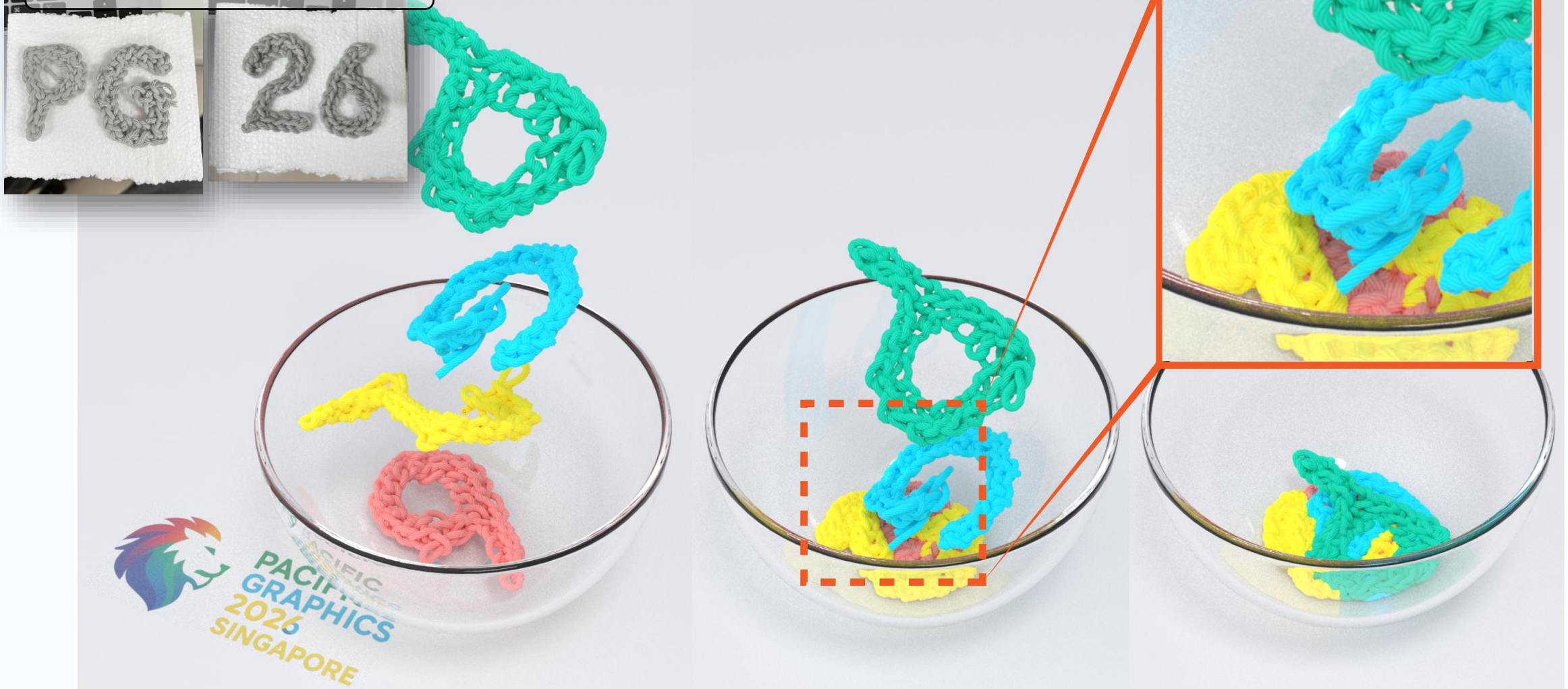
Applications: Rendering

RGB Reference Photo



Applications: Physics Simulation

RGB Reference Photo



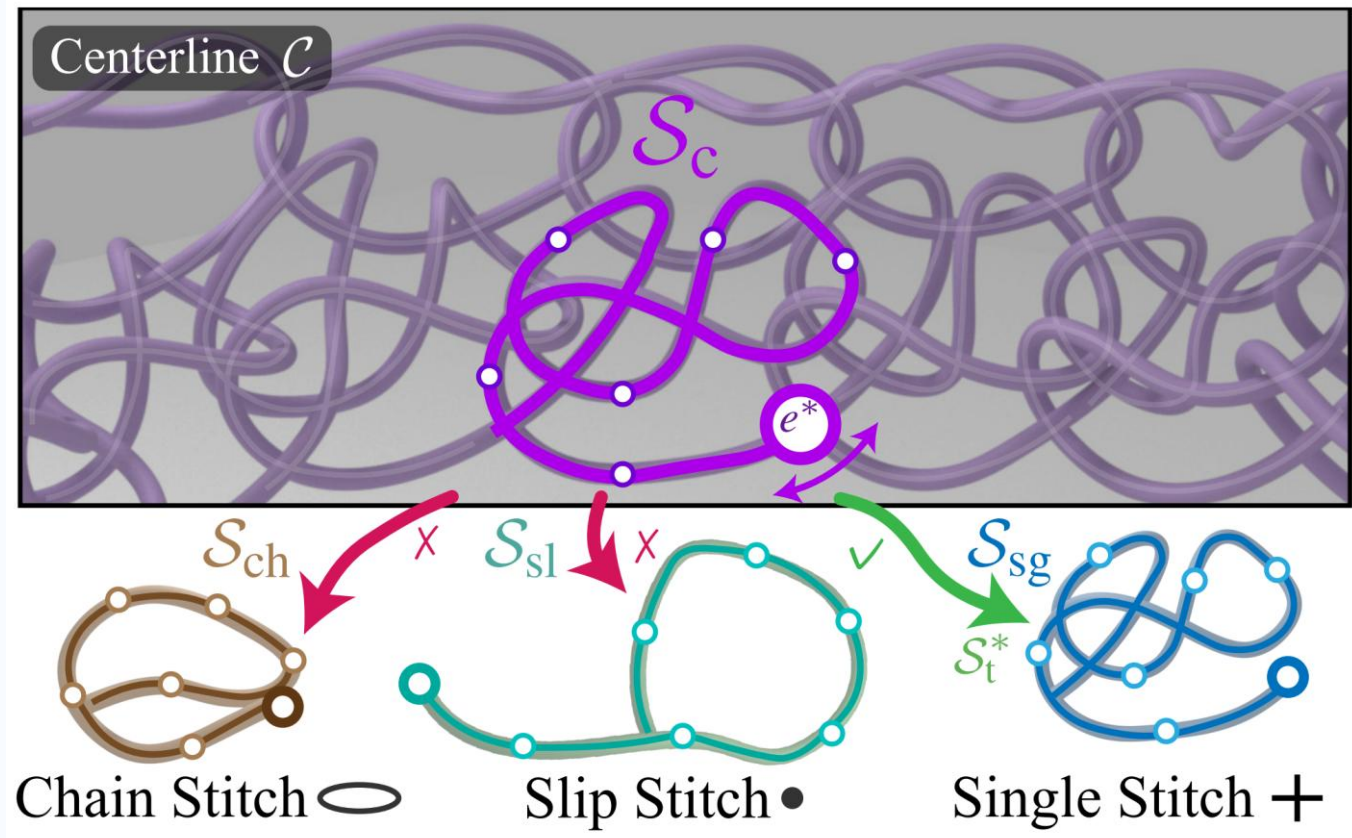
Applications: Physics Simulation

RGB Reference Photo



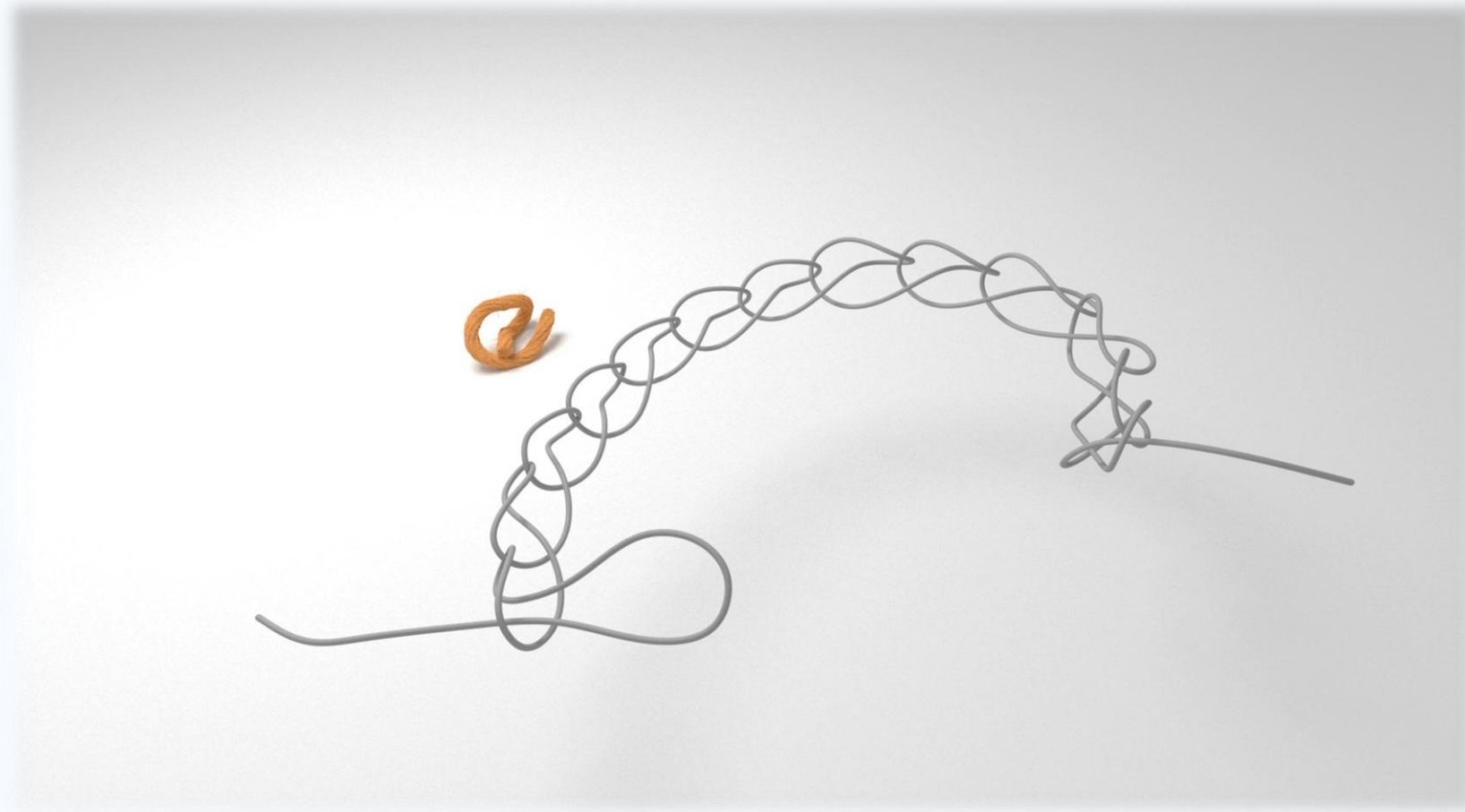
Application: Template Fitting

- Auto-regressively, can obtain the **semantic stitch pattern**!



Application: Template Fitting

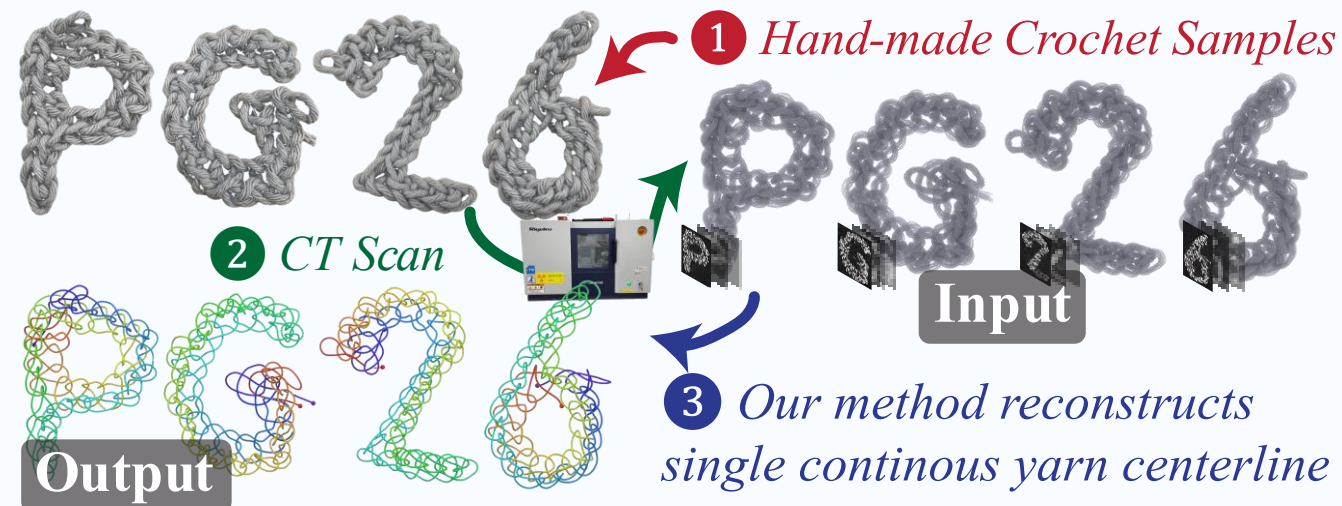
- We can obtain the **semantic stitch pattern**!



- **Human-In-the-Loop** yarn centerline reconstruction.

- **Input:** CT Scan of Crochet

- **Output:** Centerline Curve



- **Future plan:**

- Annotate more samples as a large dataset for scaling up to support the machine learning.
- Finding an optimization method to include the discrete decisions to repair the topology.



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Thank you for your attention!

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UTokyo